# Lecture 3

# Epidemic theory part 2: epidemiological and biological parameters and their interactions with transmission and interventions

Lecturer: Dr Alexandra Hogan

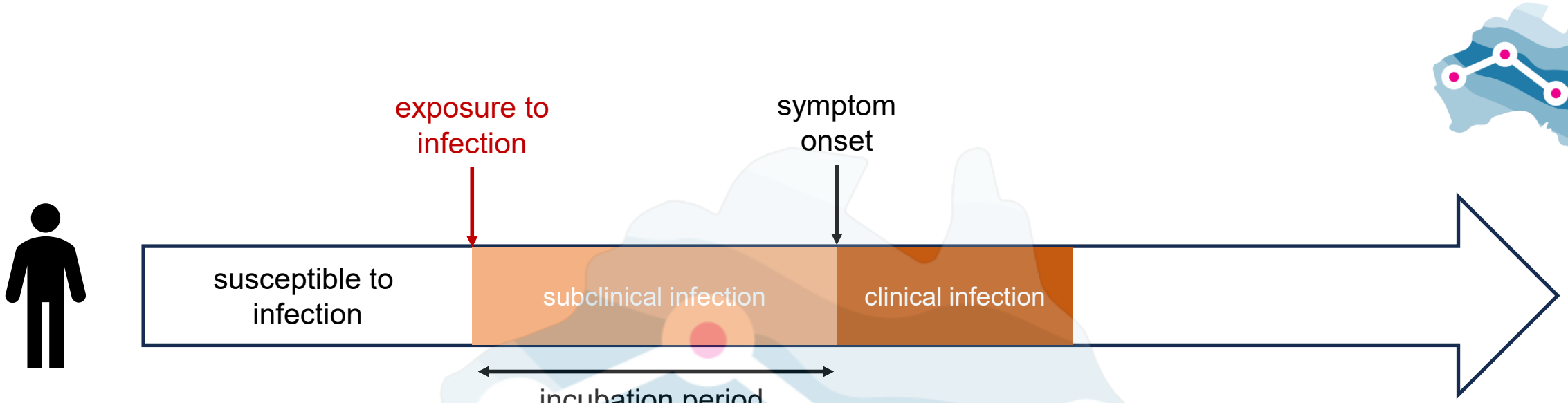
Prepared for the ACEFA EpiAnalytics Course at Australian CDC, Canberra, November 2026
Slides prepared by Alexandra Hogan, Oliver Eales, Freya Shearer

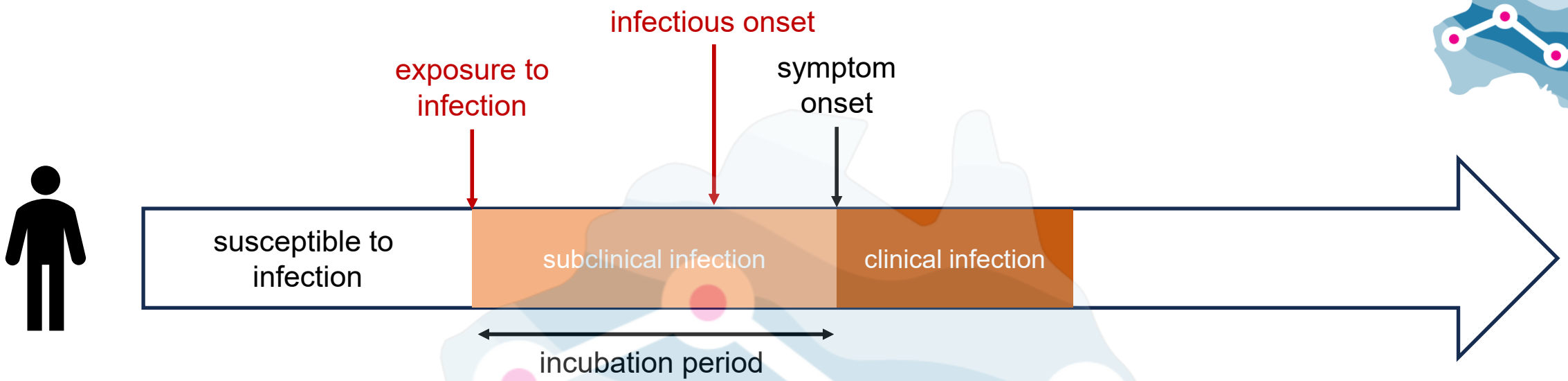
# Outline

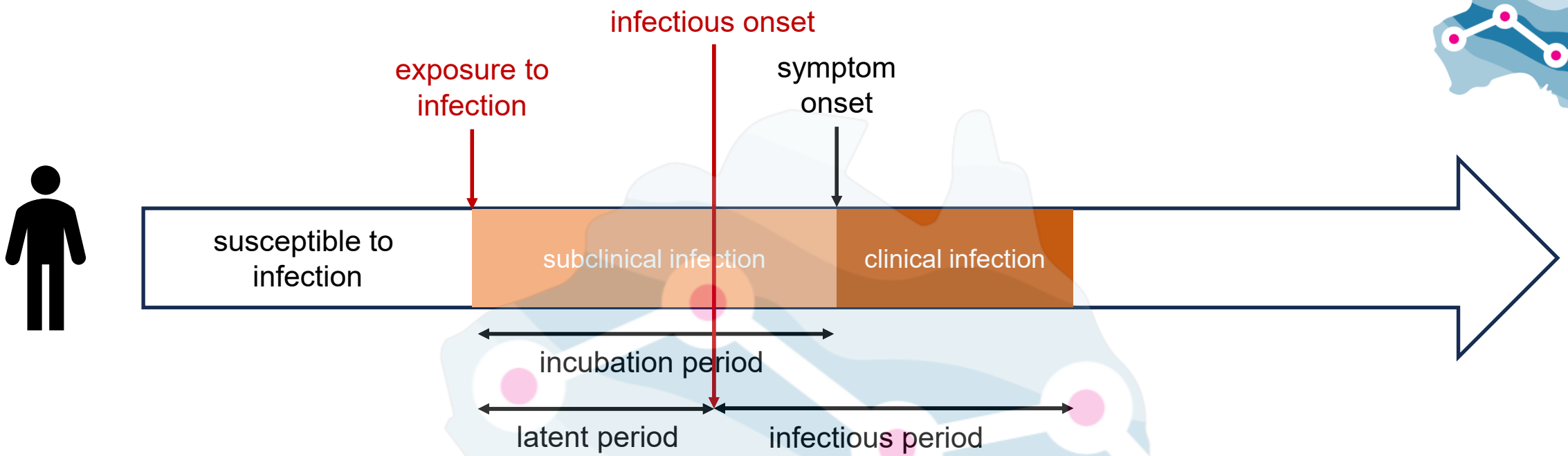
1. Key epidemiological quantities and interpretation
2. Epidemic dynamics
3. Individual-level and population-level heterogeneity
4. Modelling interventions

# Key terms that we will be exploring

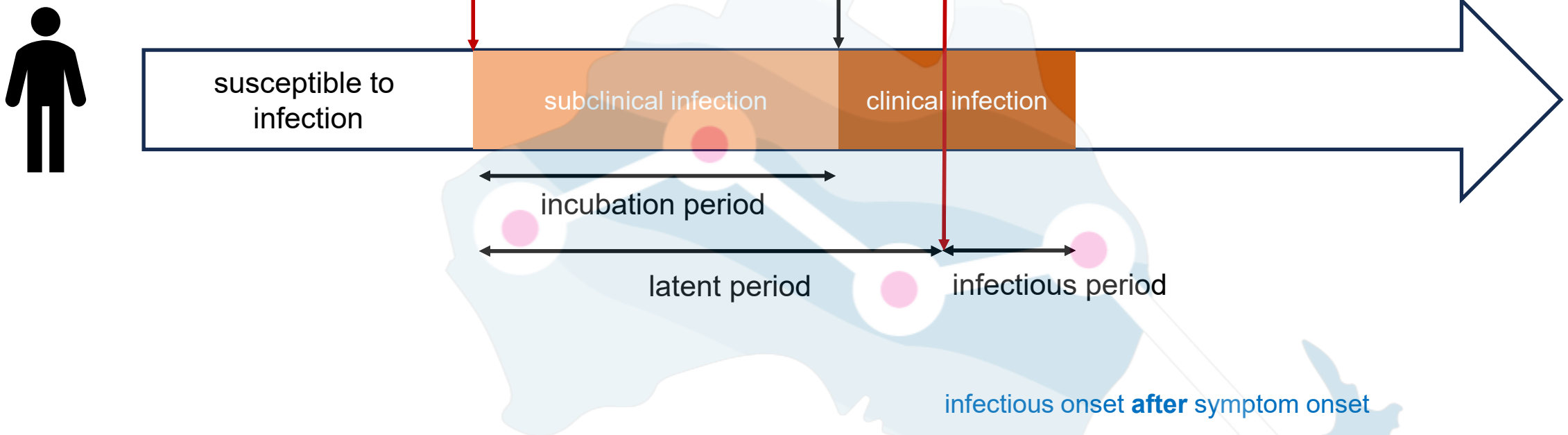
- Incubation period
- Latent period
- Infectious period
- Generation interval (or generation time)
- Serial interval


exposure to infection
symptom onset
susceptible to infection
subclinical infection
clinical infection
incubation period

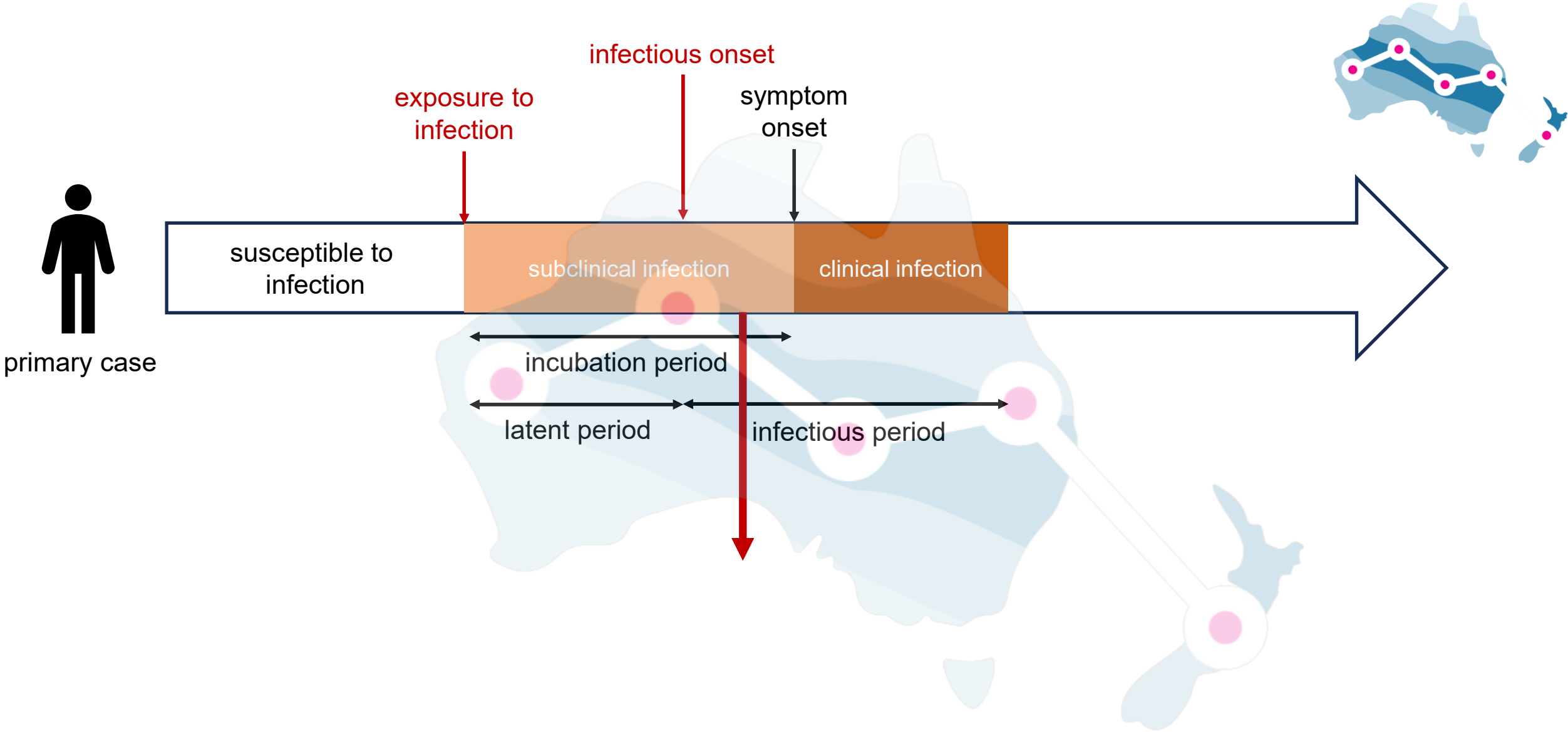

infectious onset
exposure to infection
symptom onset
susceptible to infection
subclinical infection
clinical infection
incubation period

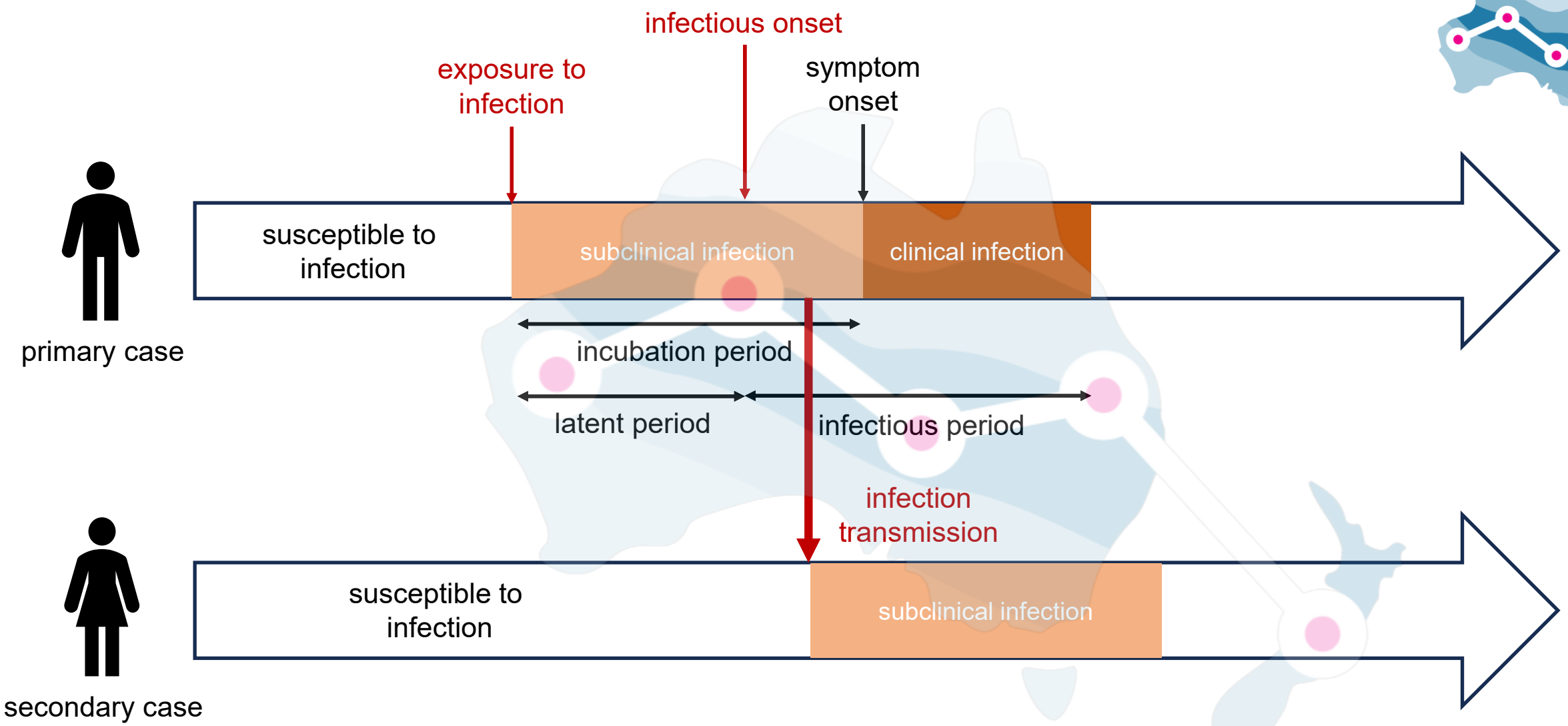


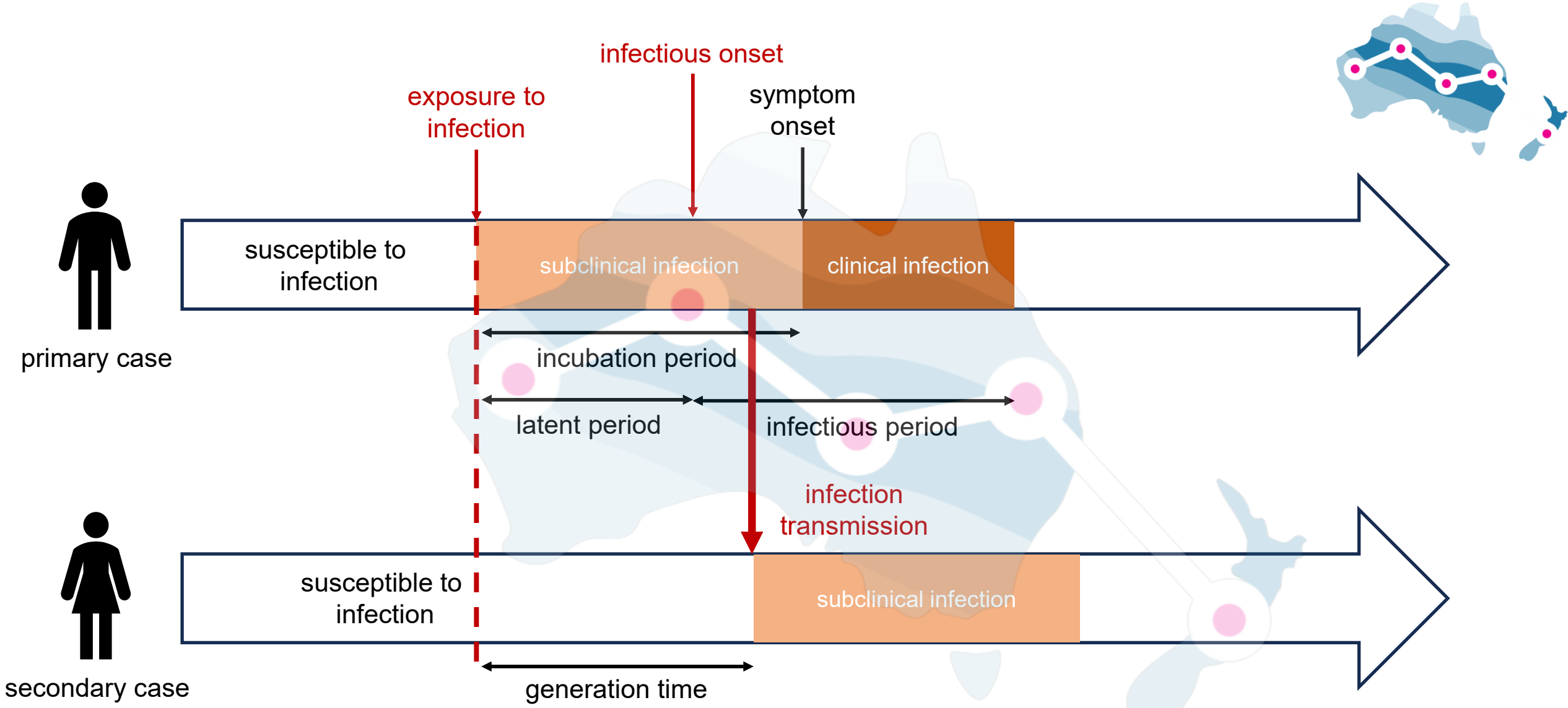

infectious onset **before** symptom onset

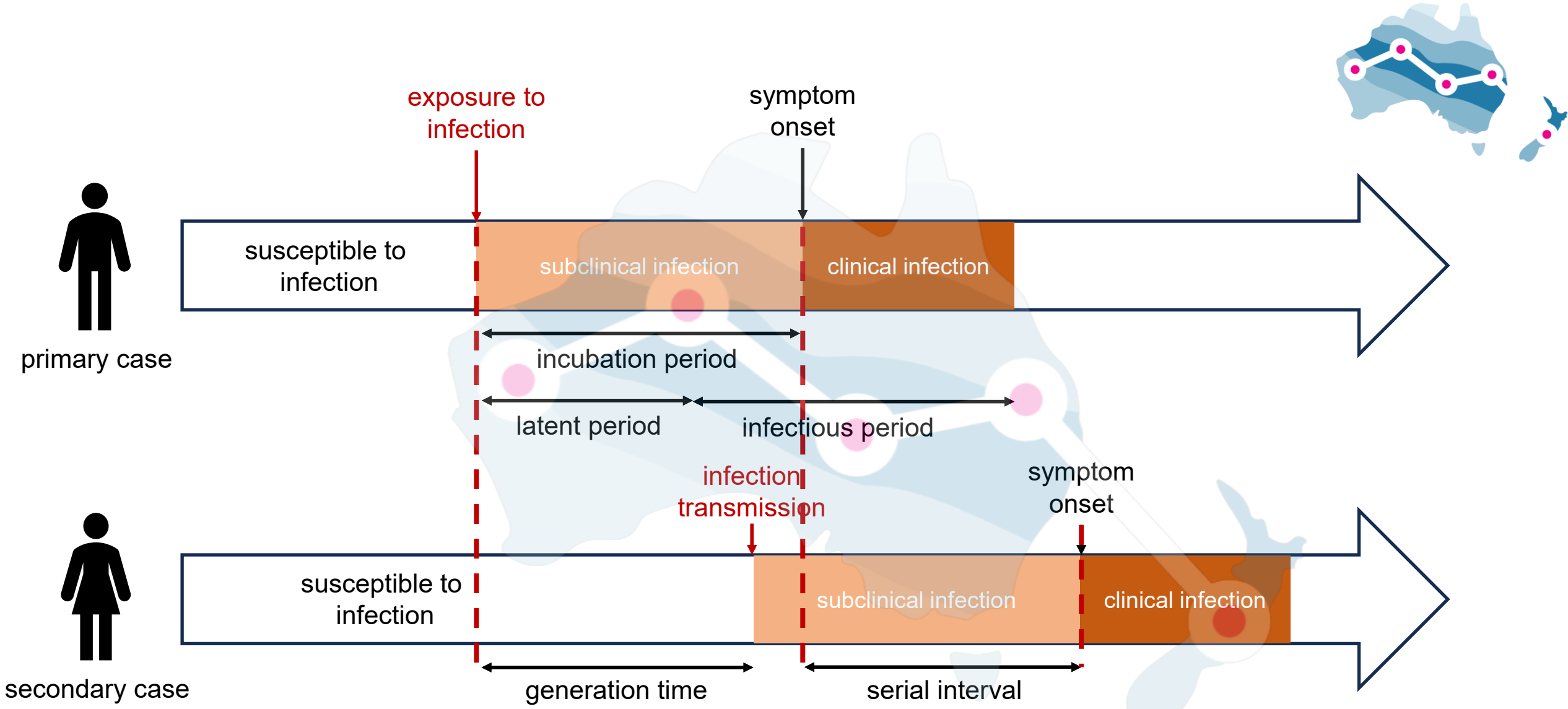



infectious onset **after** symptom onset


infectious onset
exposure to infection
symptom onset
susceptible to infection
subclinical infection
clinical infection
primary case
incubation period
latent period
infectious period


infectious onset
exposure to infection
symptom onset
susceptible to infection
subclinical infection
clinical infection
primary case
incubation period
latent period
infectious period
infection transmission
susceptible to infection
subclinical infection
secondary case

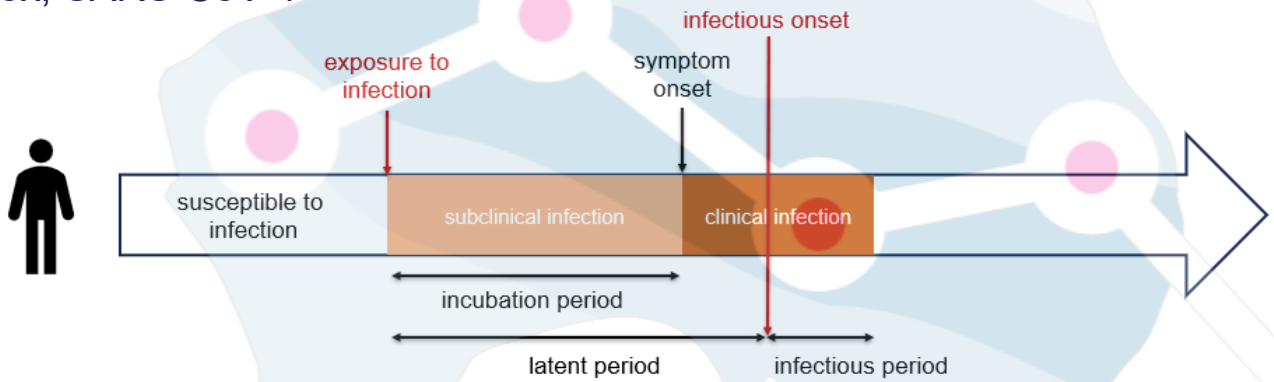

infectious onset
exposure to infection
symptom onset
susceptible to infection
subclinical infection
clinical infection
primary case
incubation period
latent period
infectious period
infection transmission
susceptible to infection
subclinical infection
secondary case
generation time


exposure to infection
symptom onset
susceptible to infection
subclinical infection
clinical infection
primary case
incubation period
latent period
infectious period
infection transmission
symptom onset
susceptible to infection
subclinical infection
clinical infection
secondary case
generation time
serial interval

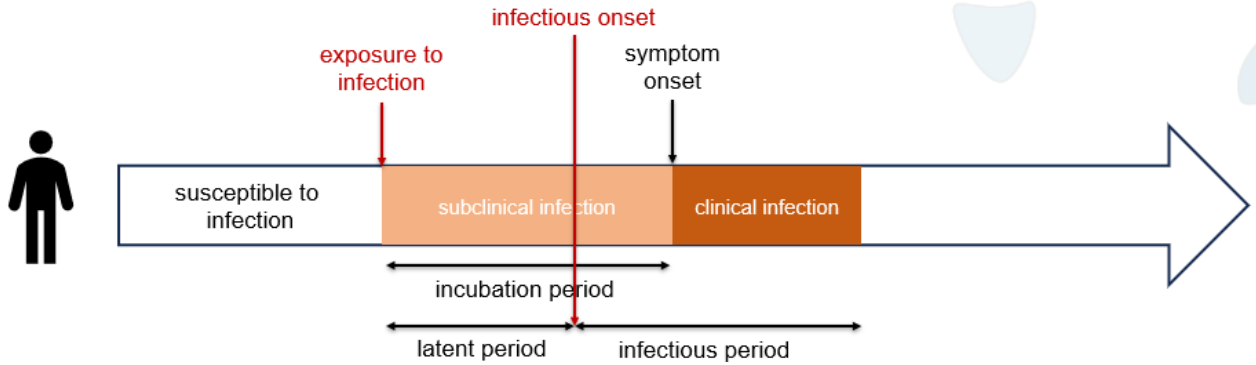
# Incubation and latent periods

The relative durations of the **incubation** and **latent** periods have implications for disease surveillance and control

- Latent period ≥ incubation period: transmission is symptomatic
  - E.g. Smallpox, SARS-CoV-1




- Latent period < incubation period: presymptomatic transmission (i.e. individual is infectious before developing symptoms)
  - E.g. SARS-CoV-2, HIV (note different timescales)



# Summary of the key terms

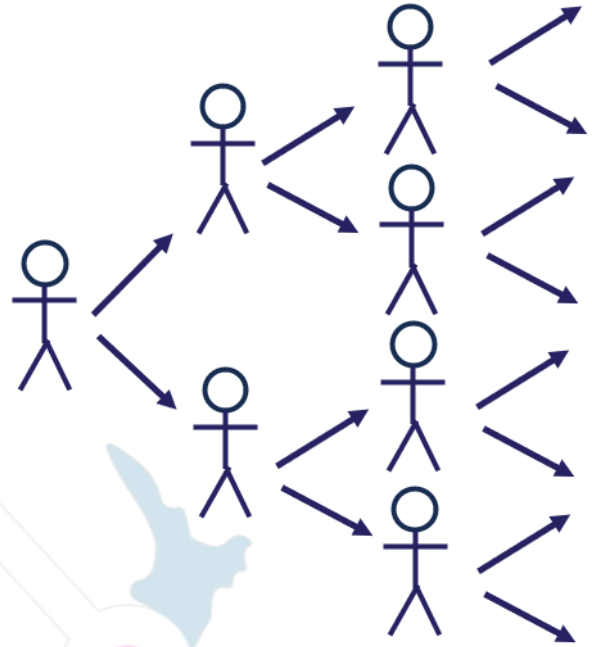
**Incubation period.** The time between exposure and onset of symptoms (within a single individual)

**Latent period.** The time between exposure and becoming infectious (within a single individual)

**Infectious period.** The duration that an individual is infectious

# Implications for $R_0$

- $R_0$ is the basic reproduction number – or number of infections generated by a single infectious individual in a fully susceptible population.
- Infectious period is important in terms of transmission dynamics
- Even if a pathogen has low transmissibility, if the infectious period is long, the $R_0$ can be high



*Note that $R_0$ does not specify the time over which infections occur*

**transmissibility**

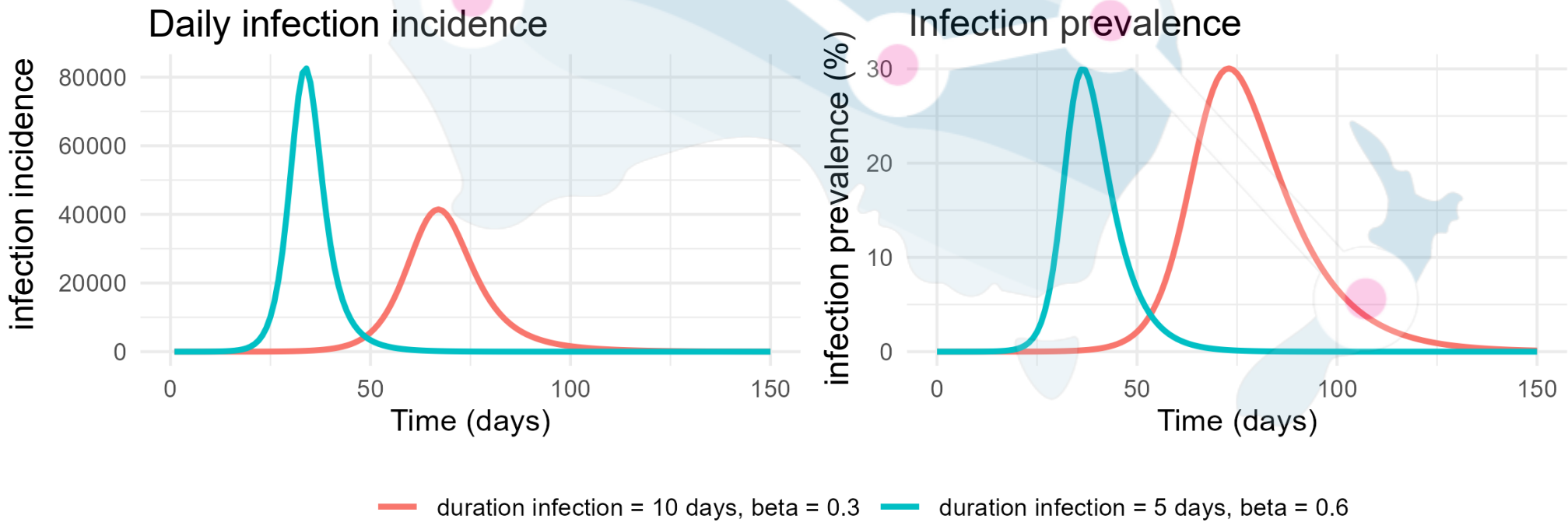
$$R_0 = \beta \times \frac{1}{\gamma}$$

**infectious period**

# Implications for $R_0$

- Remember that $\beta$ is the expected amount of people that an infectious person infects each day they are infectious
- The same $R_0$ can be comprised of a different **beta** and **duration of infection**
- These two epidemics with the same $R_0$ have different properties – **total incidence is the same**, but distributed over different time windows

**Example: SIR model with $R_0 = 3$**




$R_0$ doesn't tell us the time period over which an epidemic occurs

# Back to our list of key terms…

- Incubation period
- Latent period
- Infectious period
- **Generation interval (or generation time)**
- **Serial interval**

What about those intervals that tell us something about the timing of events between an infector-infectee pair?

# Generation interval

- The generation interval is the delay between acquisition of infection in a primary and secondary case
- Relates to how fast a pathogen is spreading (rather than $R_0$, which relates to the number of infections) and has implications for containment strategies

$$R_0 = 1 + r\mathrm{T_C}$$

mean generation interval

growth rate, or per capita change in number of new cases per unit of time

- Is a function of different factors:
  - Duration of the latent period, duration of infectiousness
  - Contact patterns, individual behaviour
- Not fixed between different infectious individuals - has a distribution

Favero et al 2022, https://pmc.ncbi.nlm.nih.gov/articles/PMC9198515/

# Estimating the generation interval

- Difficult to measure because the time of infection is not usually known
- Can sometimes be approximated by the **serial interval**, as symptom onset is easier to observe than infection (but limitations in this approach)

Chen et al 2022, https://www.nature.com/articles/s41467-022-35496-8

# Serial interval

- Time between symptom onsets in an infector–infectee pair
- In practice it is typically used to quantify $R_0$ at the start of an outbreak
- Ideally, estimated using contact tracing data (pairs of infector-infectee with known dates of symptom onset); household studies
- There are limitations of using the serial interval as an approximation of the generation time, and potential biases to account for (more in this in later lectures)[2]
- Serial intervals can vary widely between pathogens:[1]
  - Ancestral COVID-19 (4–8 days)
  - RSV (~8 days)
  - Pertussis (~20 days)
  - Smallpox (9–45 days)
  - Tuberculosis (months–years)

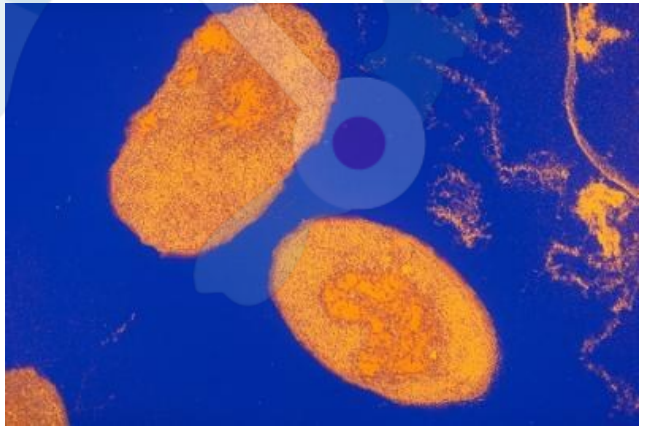


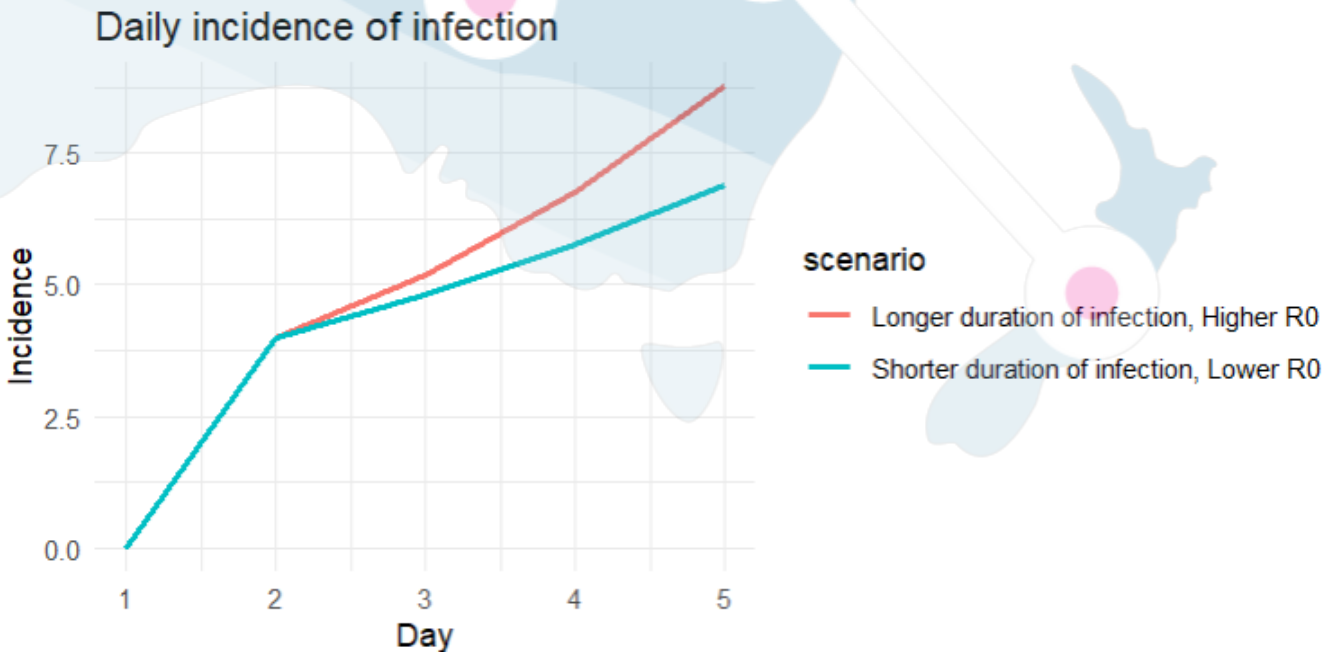
*Image:* Electron microscope image of the bacteria *Bordetella pertussis*.
*Credit: Sanofi Pasteur*
*Sourced from:* https://www.niaid.nih.gov/diseases-conditions/pertussis-whooping-cough

[1] Vink et al 2014, https://doi.org/10.1093/aje/kwu209
[2] Favero et al 2022, https://pmc.ncbi.nlm.nih.gov/articles/PMC9198515/

# How do these quantities influence epidemic dynamics?

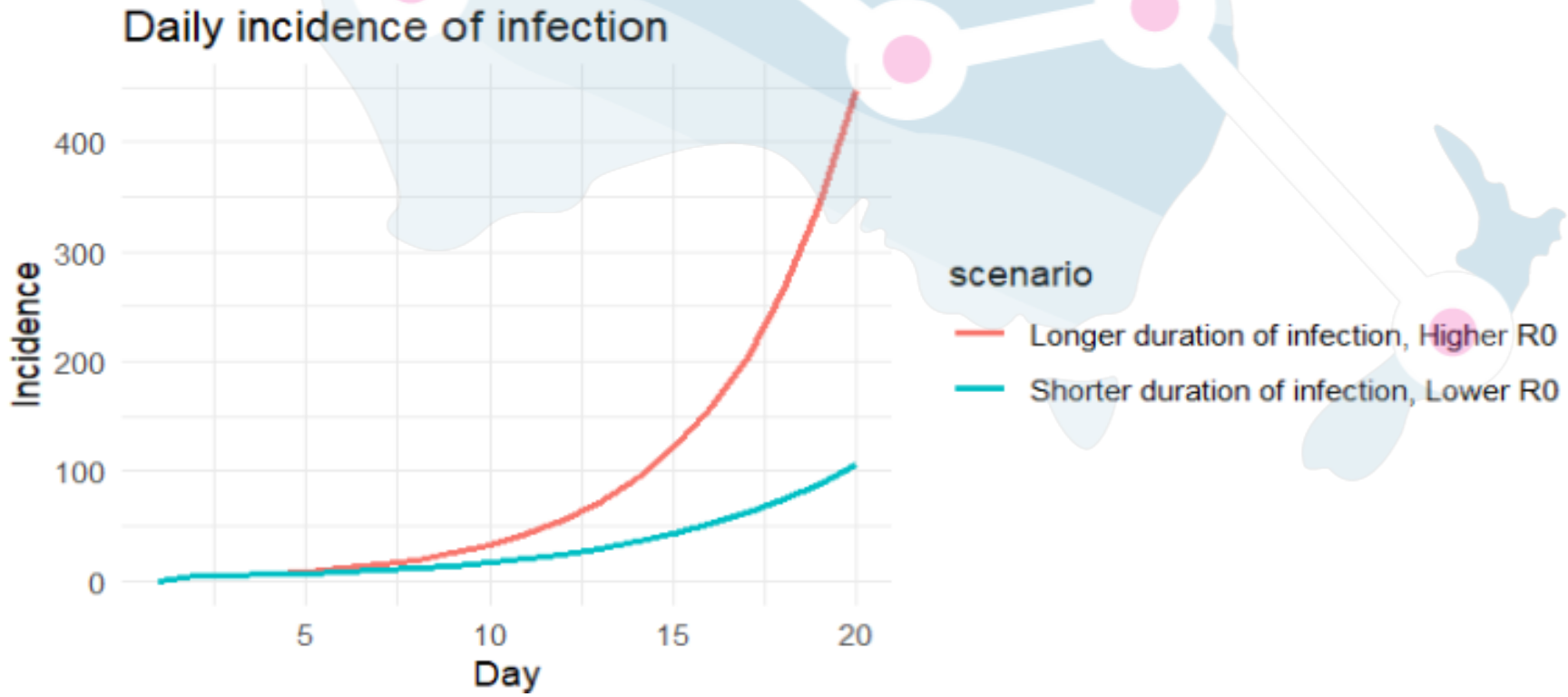
- Consider an SIR model with two different scenarios:
  - beta (the expected amount of people an infected person infects per timestep) is the same between scenarios
  - $R_0$ and the generation time are different between scenarios

In the early phase, the epidemic trajectories look similar



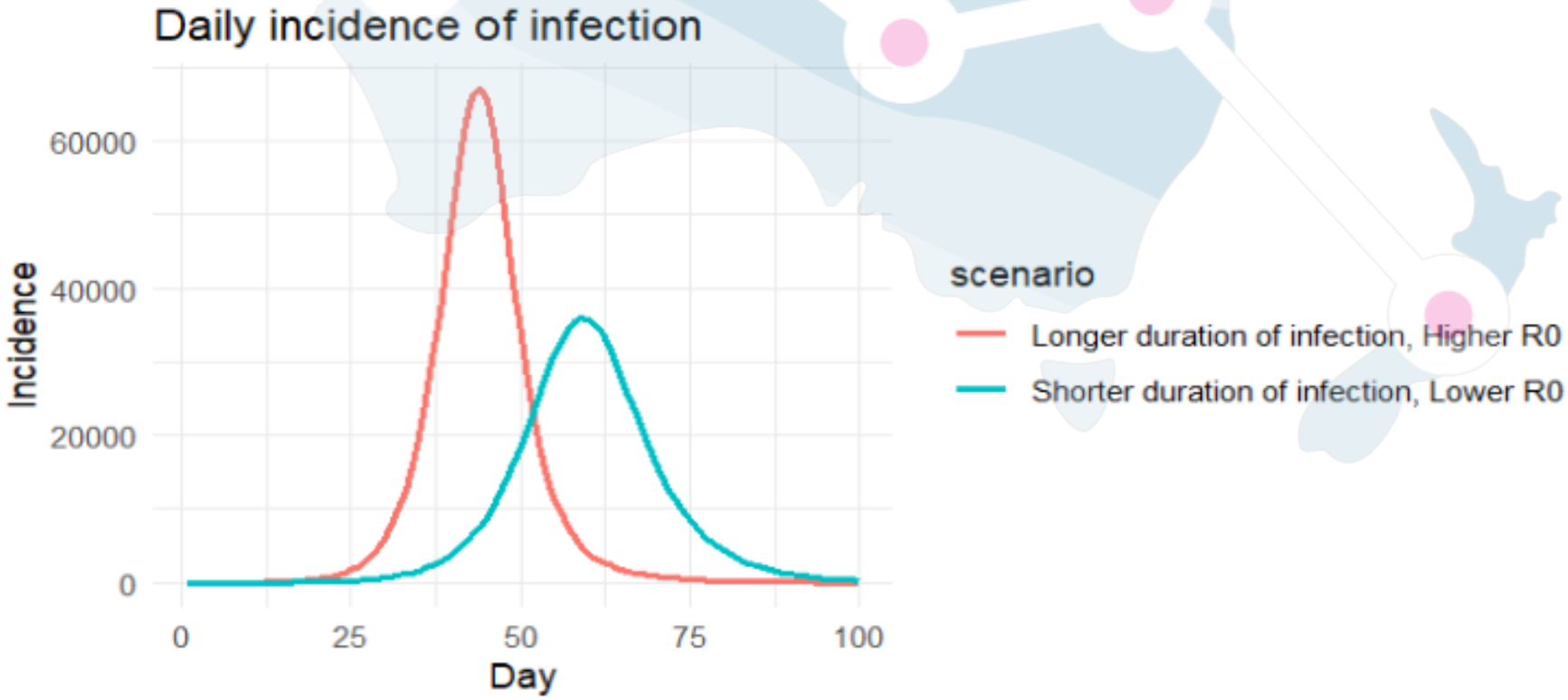
# How do these quantities influence epidemic dynamics?

- However, the epidemics then look very different and have different peak sizes and durations
- The same observed growth rate and initial case numbers can correspond to very different underlying processes (knowing the epi curve is not enough)
- This has implications for control strategies



# How do these quantities influence epidemic dynamics?

- However, the epidemics then look very different and have different peak sizes and durations
- The same observed growth rate and initial case numbers can correspond to very different underlying processes (knowing the epi curve is not enough)
- This has implications for control strategies



# Individual heterogeneity

In practice, we define these quantities across **distributions**, because

- Different individuals are infectious for different periods, or take shorter or longer times to infect others
  - Individual immune response
  - Demographic risk factors (age, sex, ethnicity)

These distributions can also change over time

- Pathogen strain



# Impact of interventions

Interventions that are applied to control an infectious disease may be designed to shorten the infectious period (and therefore also shorten the generation time), or to reduce new infections per unit time, or both

How do interventions relate to the quantities we have discussed?

- **Vaccination**: can mitigate transmission by reducing susceptibility and infectiousness (reduces growth rate)
- **Masking and social distancing**: reduces contacts, and the probability of transmission given contact (reduces growth rate)
- **Contact tracing and isolation of infectious individuals**: removes infectious individuals, reduces transmission (reduces growth rate)
- **Early case detection and isolation** (shortens infectious period)
- **Treatment** (reduces infectiousness and shortens infectious period)

# Impact of case isolation and/or contact tracing on transmission

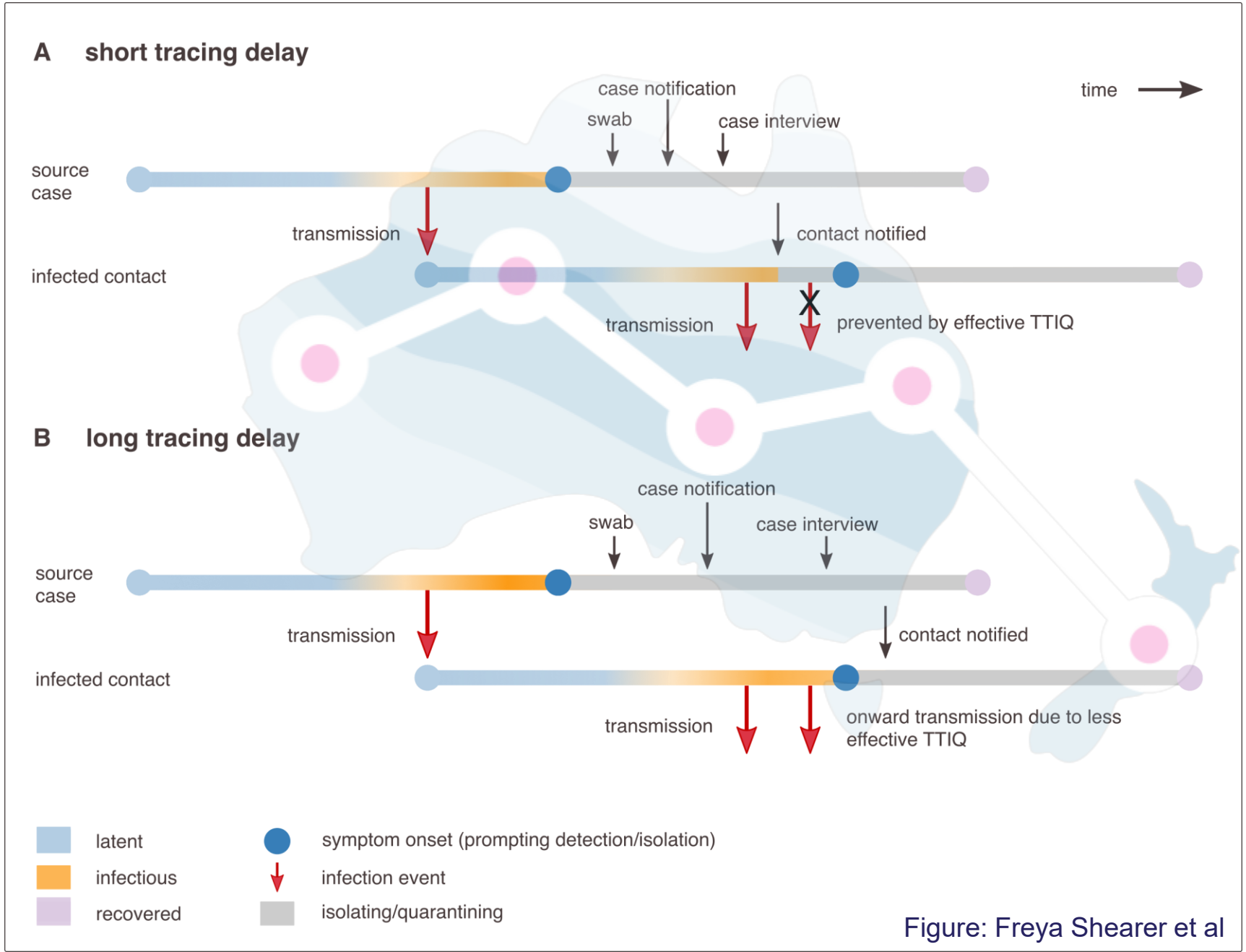
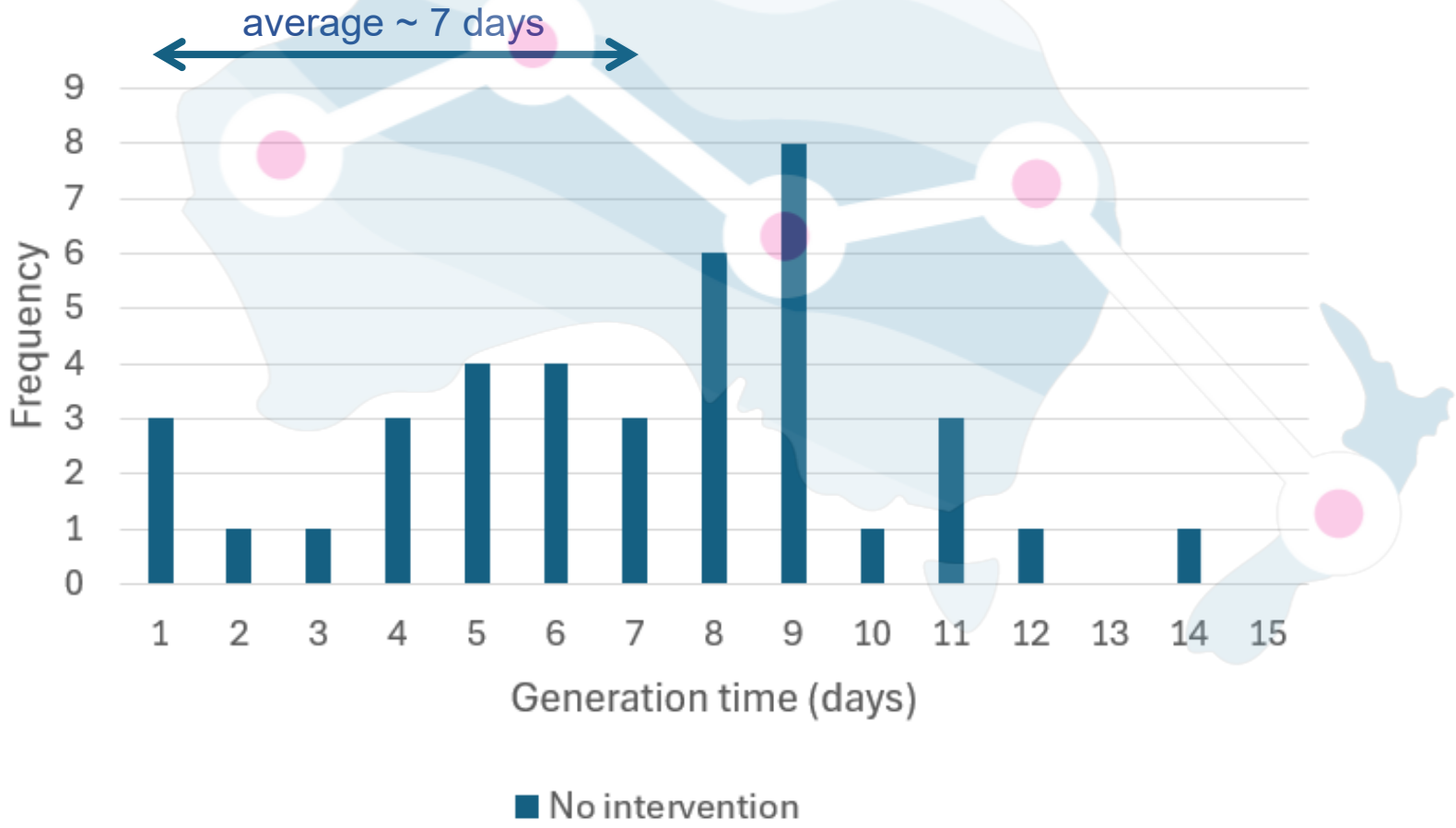



Figure: Freya Shearer et al

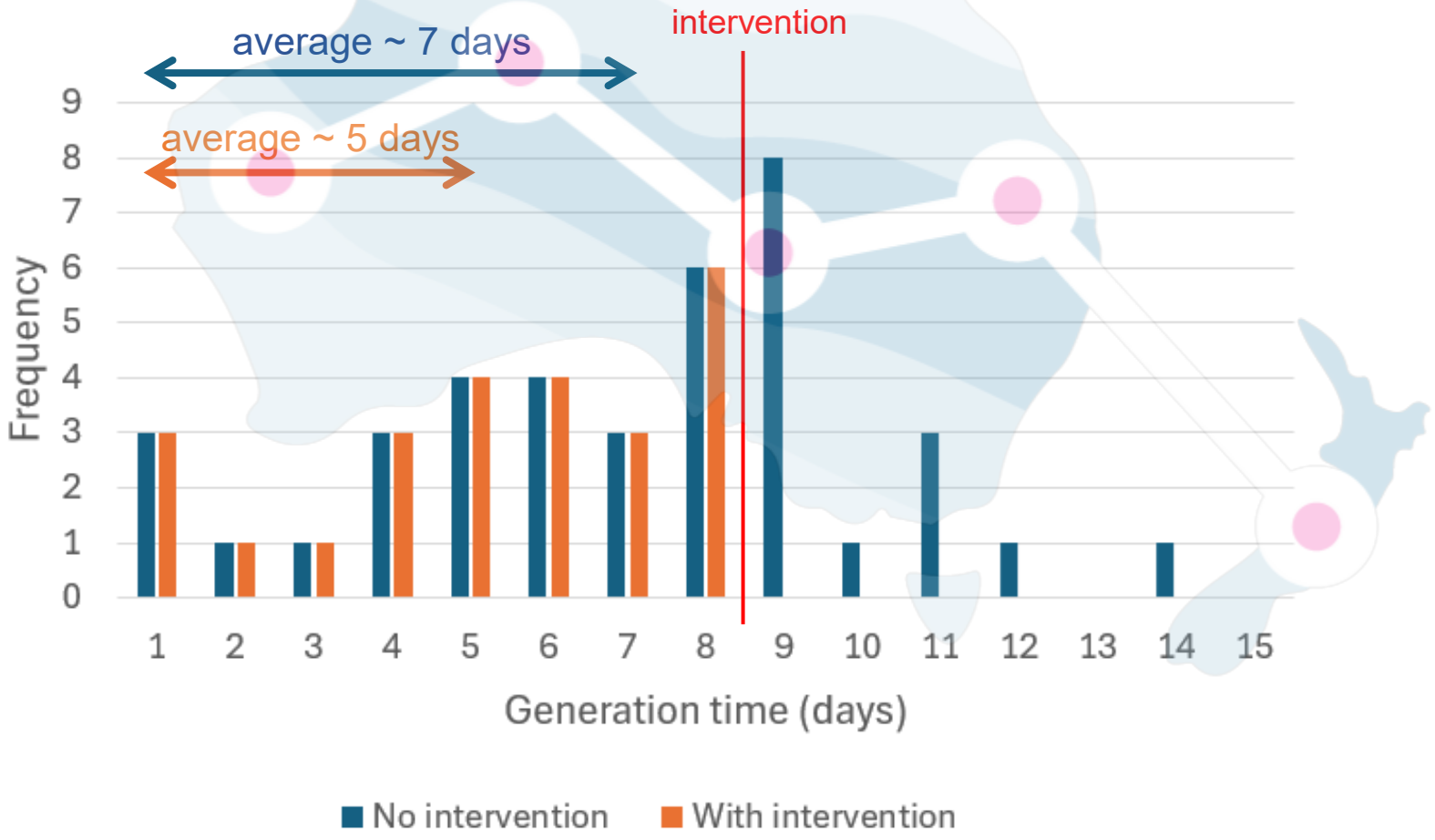
# Generation time distribution and interventions

Illustrative example of a distribution of different **generation times** for a pathogen



# Generation time distribution and interventions

If an intervention such as contact tracing or case isolation is introduced, the longer generation times are not observed



# Using modelling to explore the impact of interventions

- A model gives us a starting point to examine the impact of a new intervention, such as a new or modified vaccine program
- We typically start with the SIR model, and include different compartments and features to capture vaccination
- The specific structure depends on the disease and vaccine being modelled
- We can then compare different rollout strategies, or assumptions about vaccine characteristics and community acceptance
- These outputs can then help to inform decision-making
- Range of different ways that modelled outputs are presented



# Using modelling to explore the impact of interventions

Short-term and long-term horizons

- Prediction: “what do we think is going to happen next?”
- Scenarios: “what will happen if we do X?”

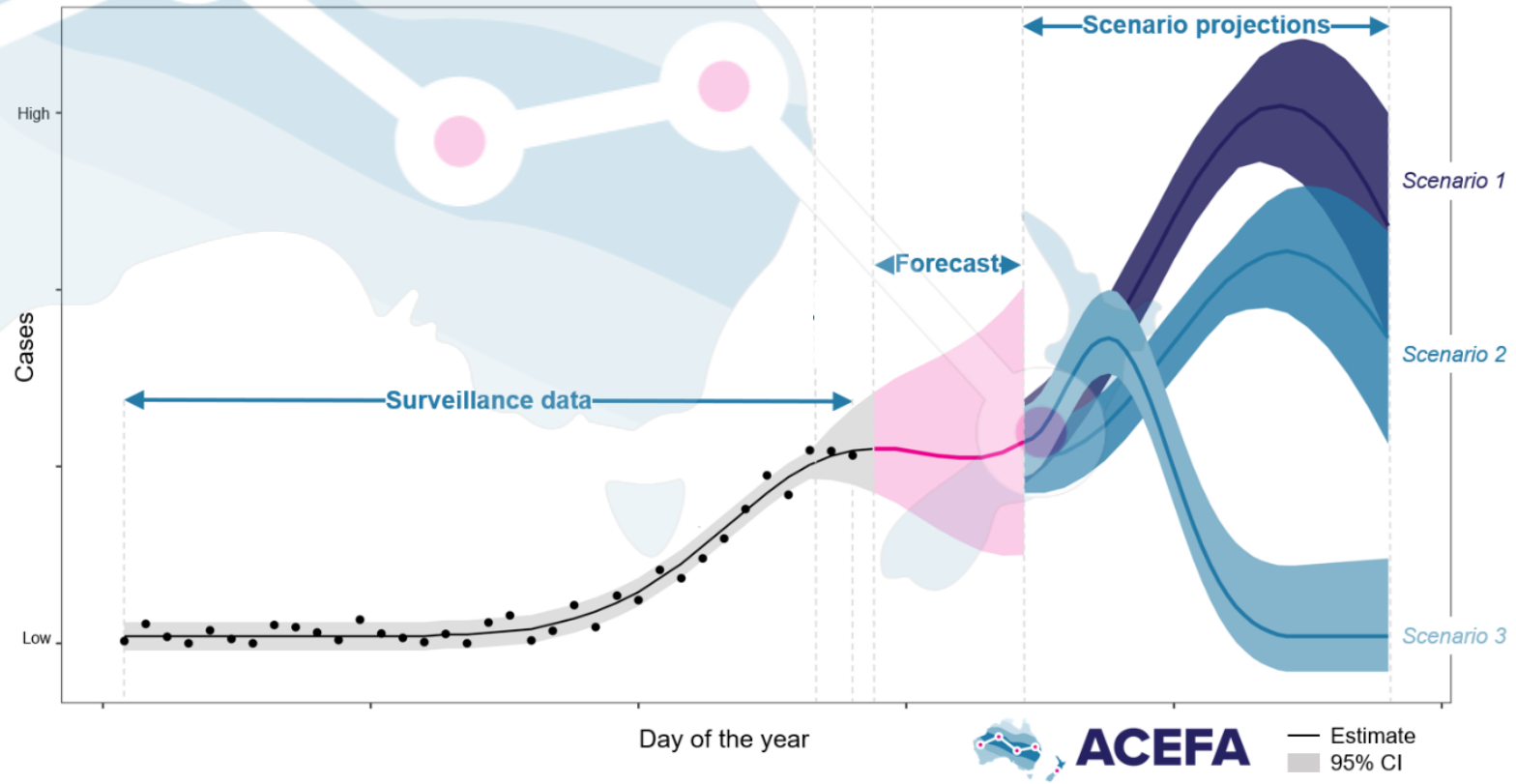
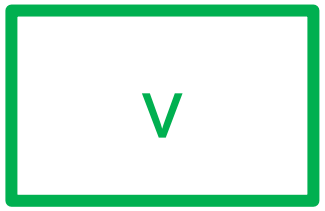



Figure: https://acefa-epi-analytics.org/activities/decision-support.html

# Using modelling to explore the impact of interventions

Example: a proportion of the population is removed to a Vaccinated (i.e. immune) class with **complete protection**




$$\frac{ds(t)}{dt} = -\beta s(t) i(t)$$

$$\frac{di(t)}{dt} = \beta s(t) i(t) - \gamma i(t)$$

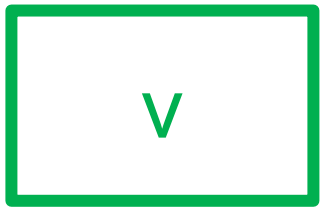
$$\frac{dr(t)}{dt} = \gamma i(t)$$

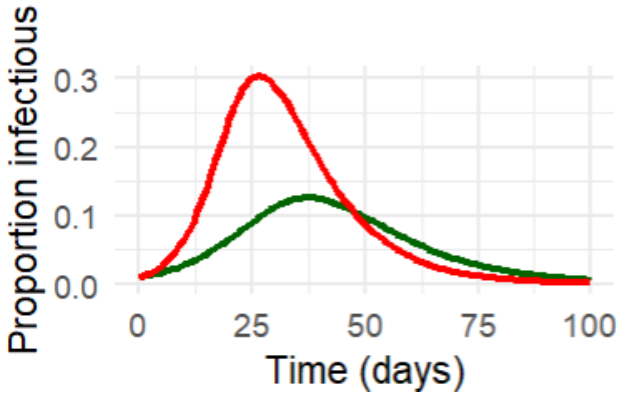
$$s(t) + i(t) + r(t) + v(t) = 1$$

# Using modelling to explore the impact of interventions

Example: a proportion of the population is removed to a Vaccinated (i.e. immune) class with **complete protection**




Scenario
- 30% Vaccination
- No Vaccination

$$\frac{d\,s(t)}{dt} = -\beta s(t) i(t)$$

$$\frac{di(t)}{dt} = \beta s(t) i(t) - \gamma i(t)$$

$$\frac{dr(t)}{dt} = \gamma i(t)$$

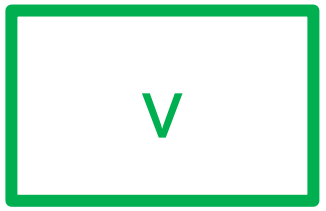
$$s(t) + i(t) + r(t) + v(t) = 1$$

Note that initial conditions (i.e. values of S, I, R and V at time 0) would be set to capture vaccine coverage

# Using modelling to explore the impact of interventions

Example: a proportion of the population is removed to a Vaccinated (i.e. immune) class with **complete protection**




$$\frac{ds(t)}{dt} = -\beta s(t) i(t)$$

$$\frac{di(t)}{dt} = \beta s(t) i(t) - \gamma i(t)$$

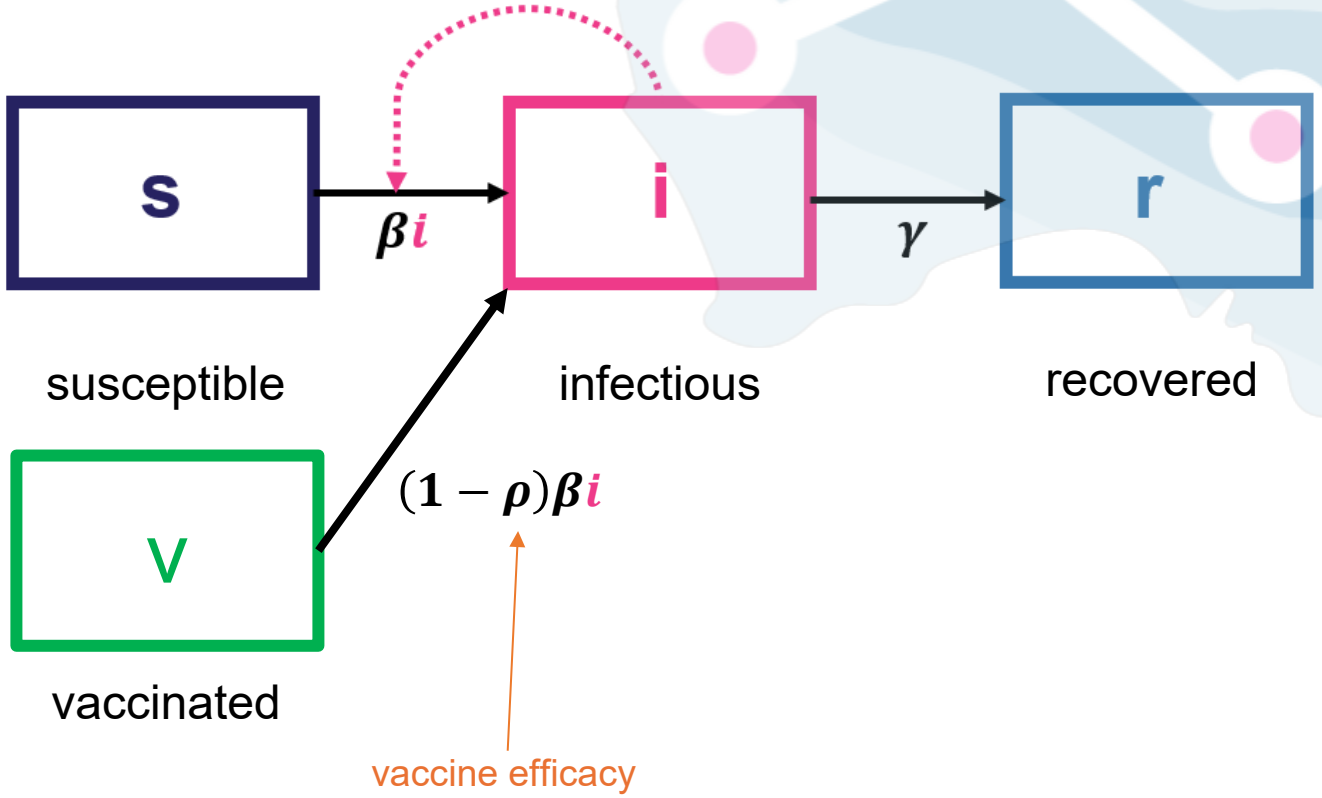
$$\frac{dr(t)}{dt} = \gamma i(t)$$

$$s(t) + i(t) + r(t) + v(t) = 1$$

Note that initial conditions (i.e. values of S, I, R and V at time 0) would be set to capture vaccine coverage

# Using modelling to explore the impact of interventions

Example: a proportion of the population is removed to a Vaccinated (i.e. immune) class with **complete protection**




$$\frac{ds(t)}{dt} = -\beta s(t) i(t)$$

$$\frac{di(t)}{dt} = \beta s(t) i(t) - \gamma i(t)$$

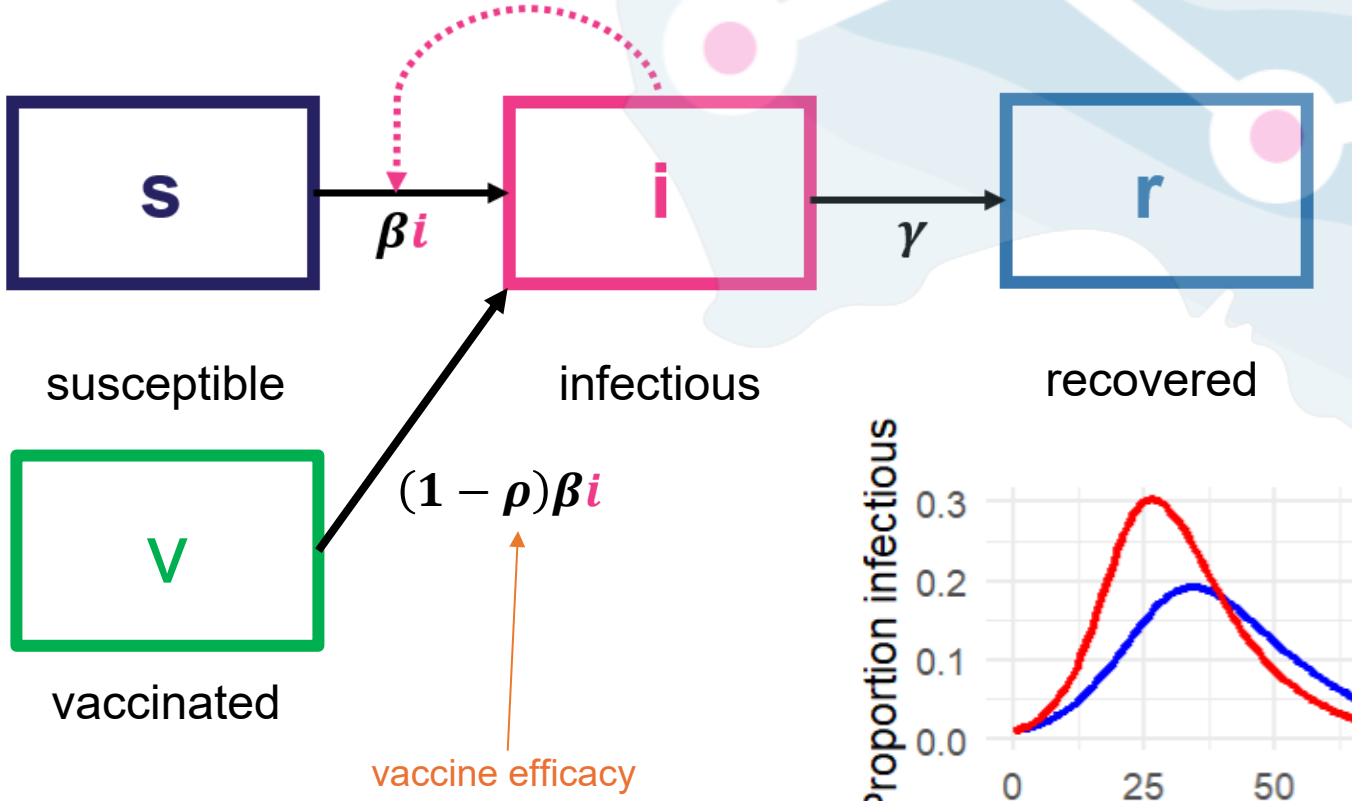
$$\frac{dr(t)}{dt} = \gamma i(t)$$

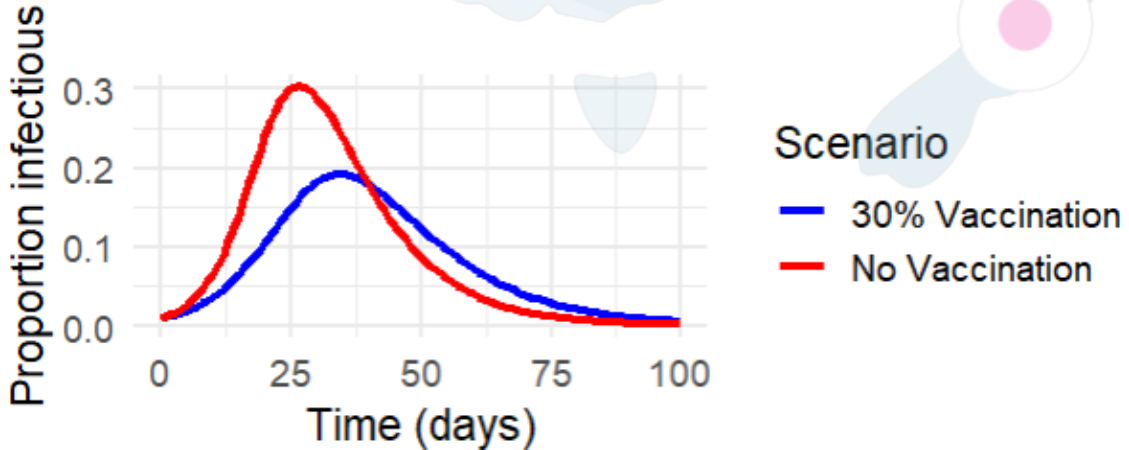
$$\frac{dv(t)}{dt} = (1-\rho)\beta v(t) i(t)$$

$$s(t) + i(t) + r(t) + v(t) = 1$$

# Using modelling to explore the impact of interventions

Example: a proportion of the population is removed to a Vaccinated (i.e. immune) class with **complete protection**



# Population heterogeneity



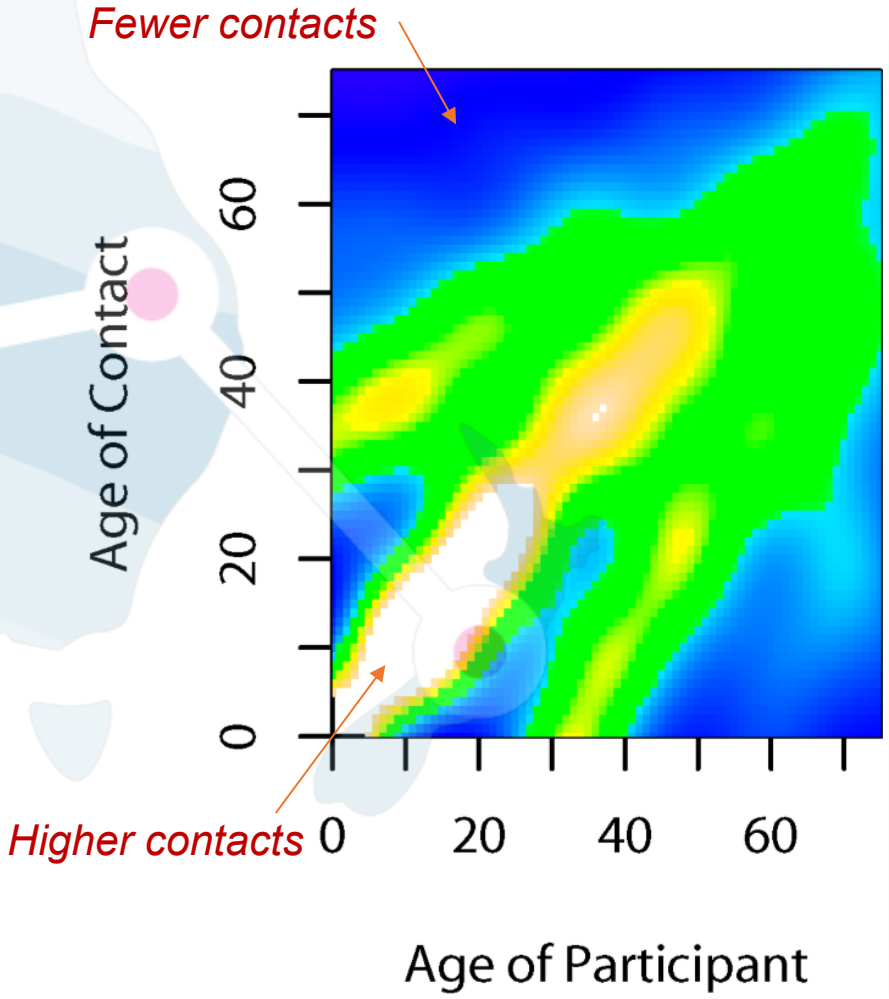
Models can be further adapted to account for different population groups

The important groups might depend on

- Disease being modelled
- Policy question being asked
- Data available to inform parameters

Age the most important factor that is typically included (by additional states in an SIR model)

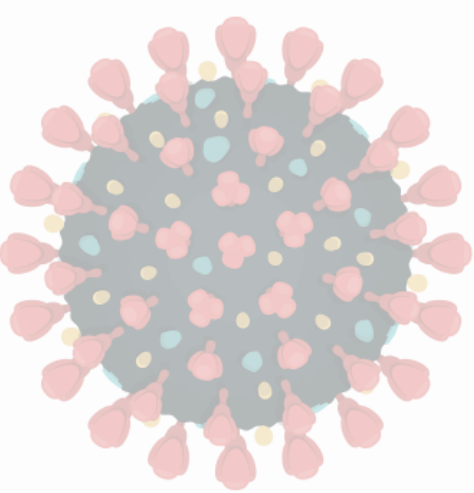
Social mixing within and between age groups captured by contact matrices[1]





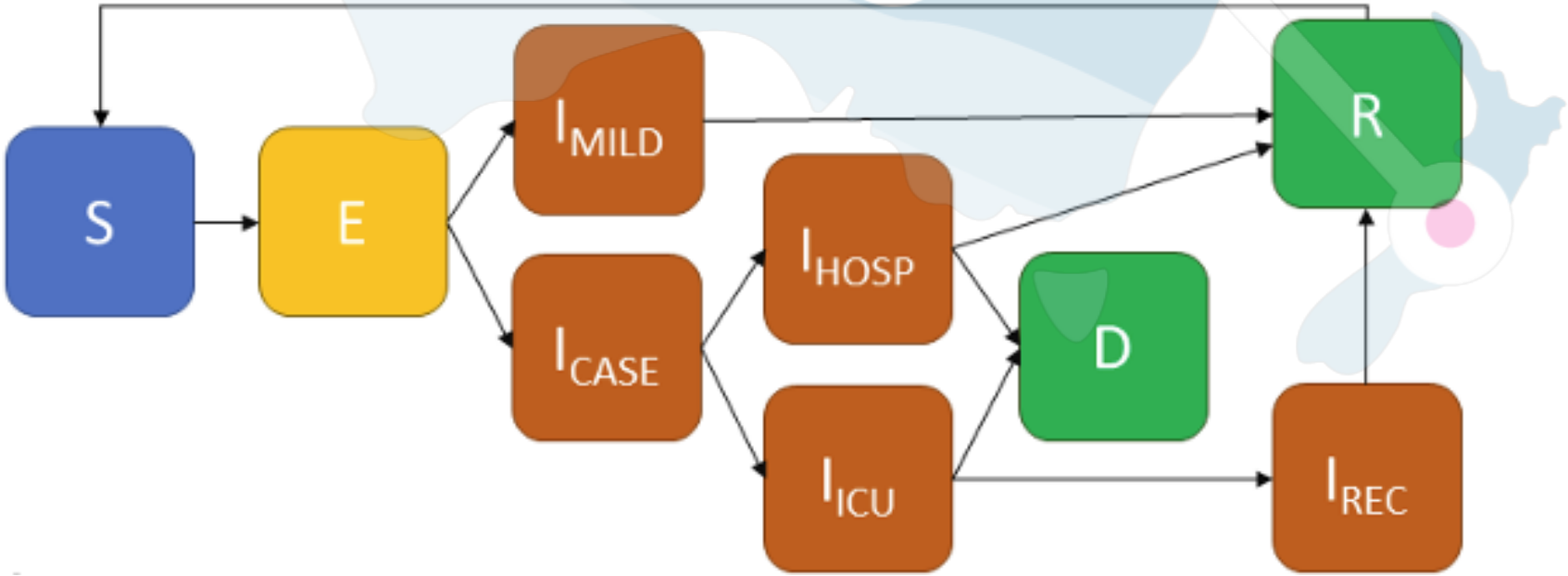

[1] eg Mossong et al 2008, https://doi.org/10.1371/journal.pmed.0050074

# Example: early SARS-CoV-2 modelling



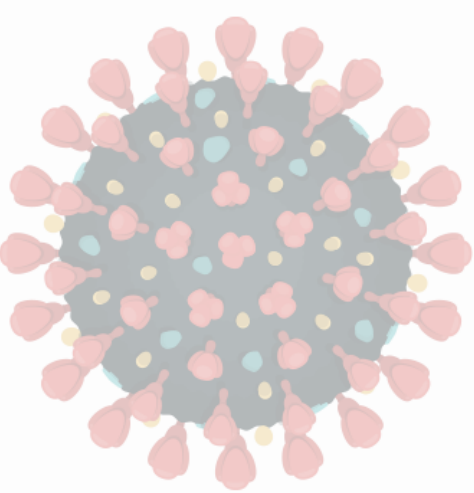
Simple SIR models such as these used early in COVID-19 pandemic to:

- Understand epidemiology
- Forecast
- Predict impact of vaccination




Imperial College London

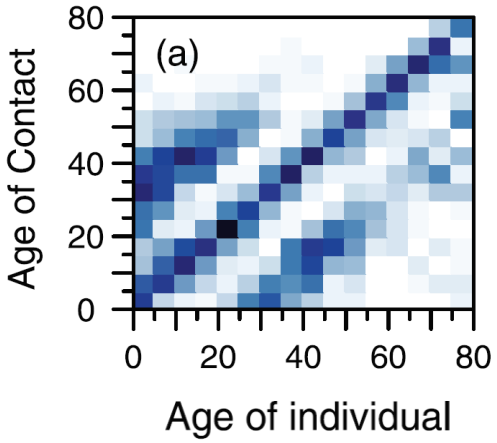
# Example: early SARS-CoV-2 modelling



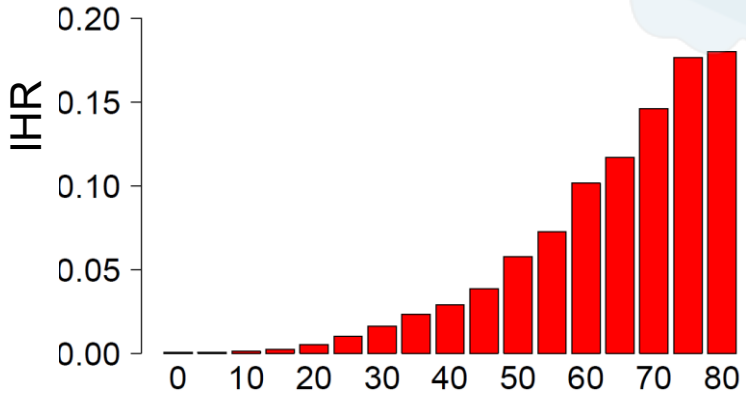
Simple SIR models such as these used early in COVID-19 pandemic to:

- Understand epidemiology
- Forecast
- Predict impact of vaccination

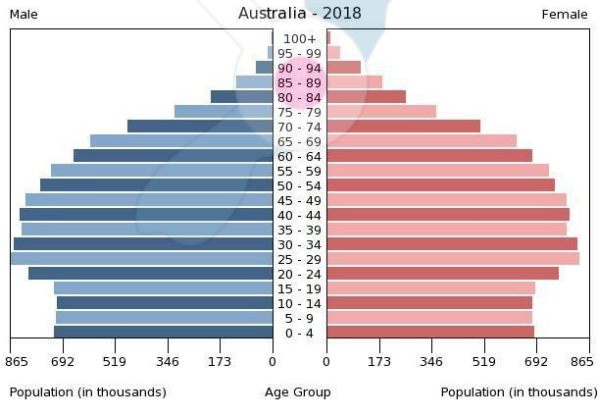
Country-specific patterns of mixing between age groups



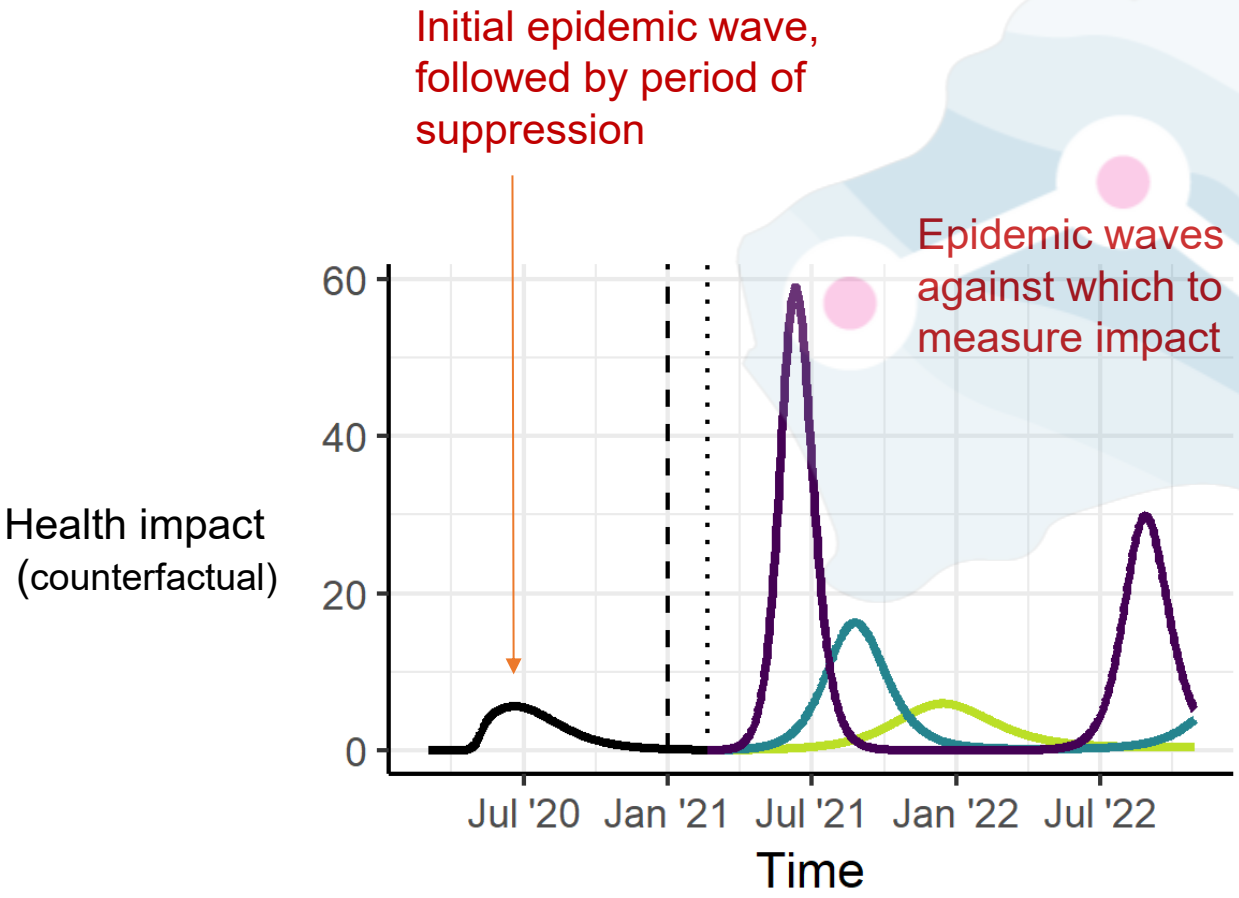

Age-dependent patterns of disease severity




Age structure (UN World Population Prospects)



# Example: early SARS-CoV-2 modelling




Coloured lines: different scenarios for how $R_0$ varied over time

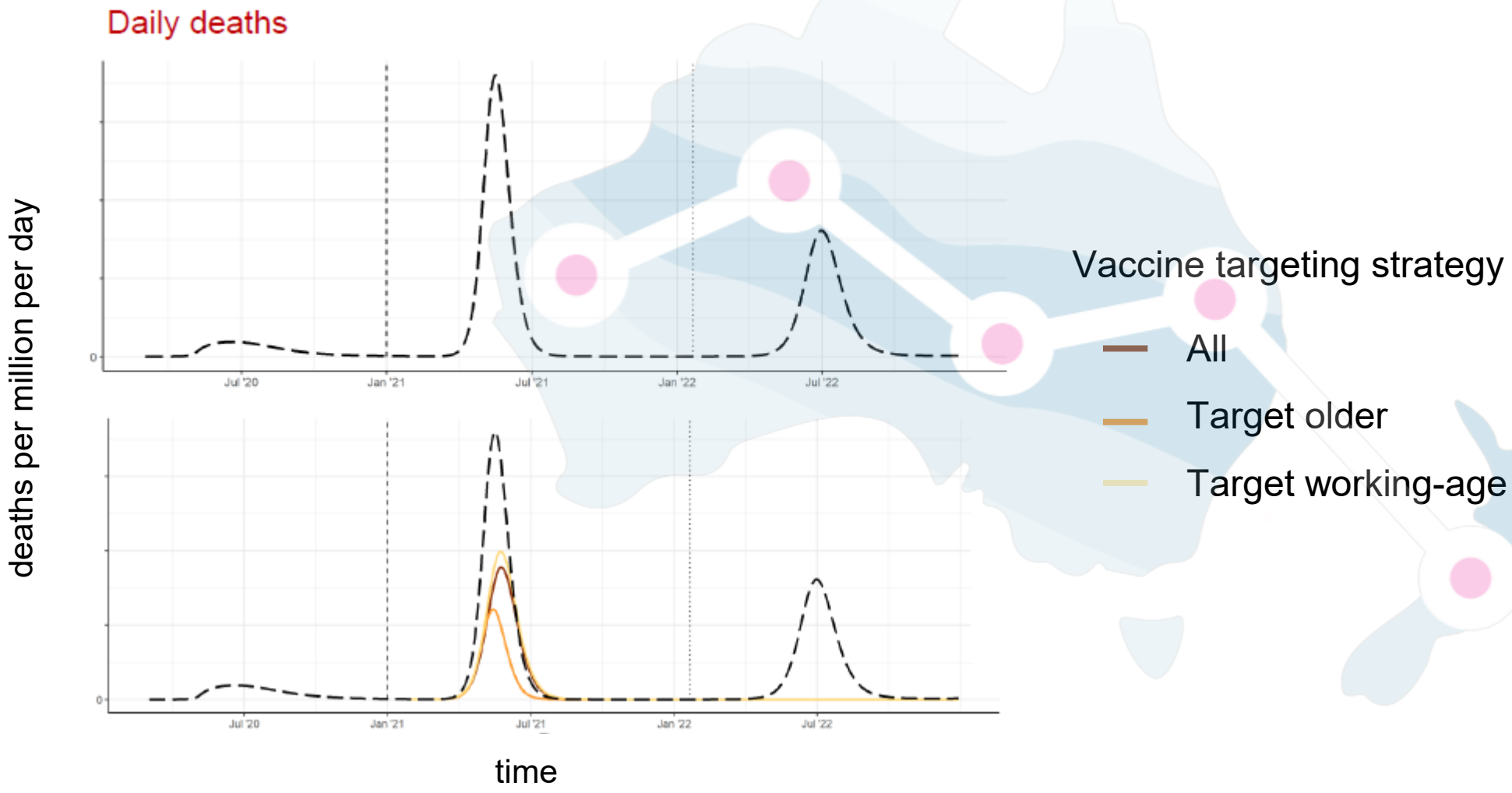
Can use these scenarios to quantify impact of vaccination

- How many deaths would a vaccine avert?
- Who to vaccinate?
- Timing of lifting lockdowns?
- Unknown vaccine characteristics and supply?

Hogan & Winskill, Vaccine 2021, https://www.sciencedirect.com/science/article/pii/S0264410X21004278#s0070

Imperial College London

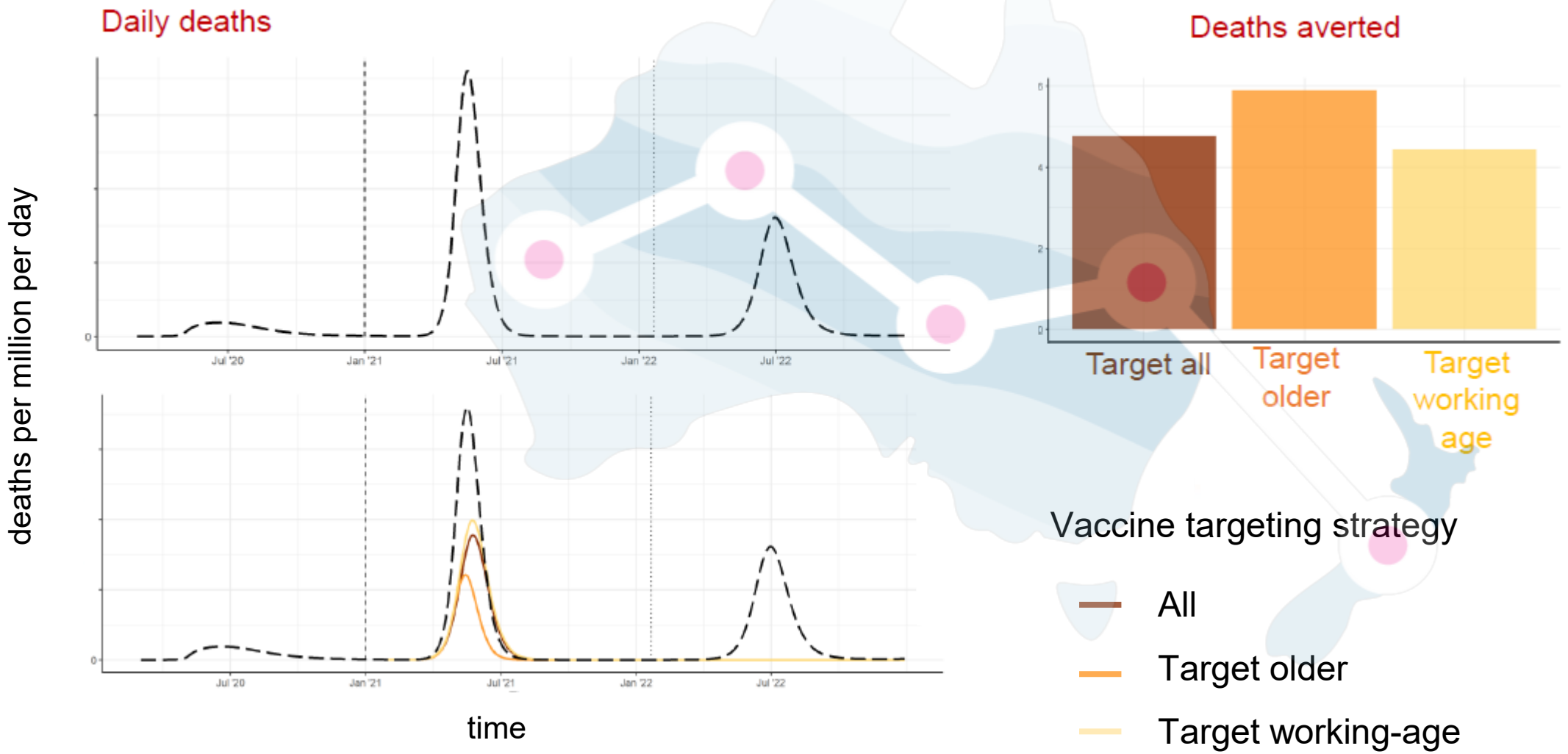
# Example: early SARS-CoV-2 modelling (Broad global high-income country setting)




Hogan & Winskill, Vaccine 2021, https://www.sciencedirect.com/science/article/pii/S0264410X21004278#s0070

Imperial College London

# Example: early SARS-CoV-2 modelling (Broad global high-income country setting)

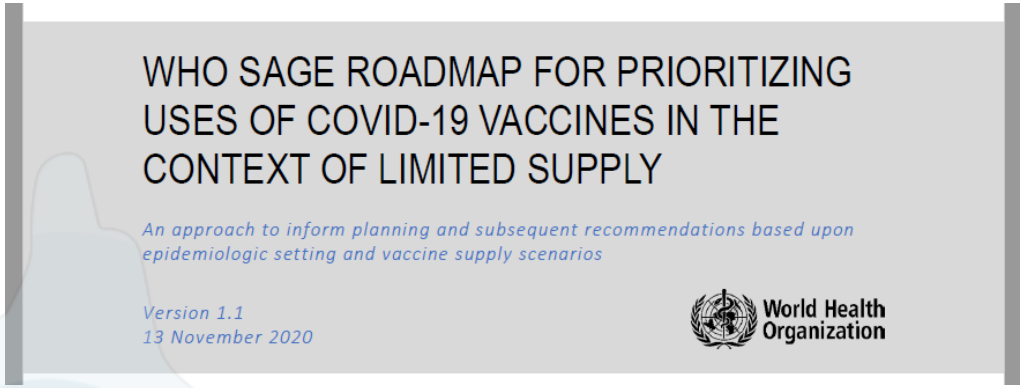


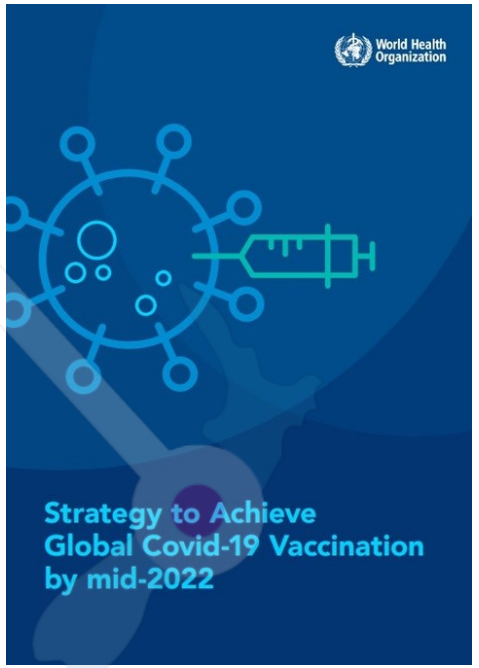

Hogan & Winskill, Vaccine 2021, https://www.sciencedirect.com/science/article/pii/S0264410X21004278#s0070

Imperial College London

# Translation to policy

- Part of a larger body of COVID-19 vaccine prioritisation work that contributed to evidence base for global health policy guidance on COVID-19 vaccination
- “WHO SAGE Roadmap for Prioritizing Uses of COVID-19 Vaccines in the Context of Limited Supply” (Nov 2020, subsequently updated)
- “WHO Strategy to Achieve Global Govid-19 Vaccination by mid-2022” (Oct 2021)




Imperial College London

# Herd immunity

- Herd immunity is an important form of indirect protection that applies to infectious diseases
- It occurs when there are enough immune individuals in a population such that each infectious individual can no longer infect more than one person
- This threshold is directly related to the reproduction number $R_0$

$$p_c = 1 - \frac{1}{R_0}$$

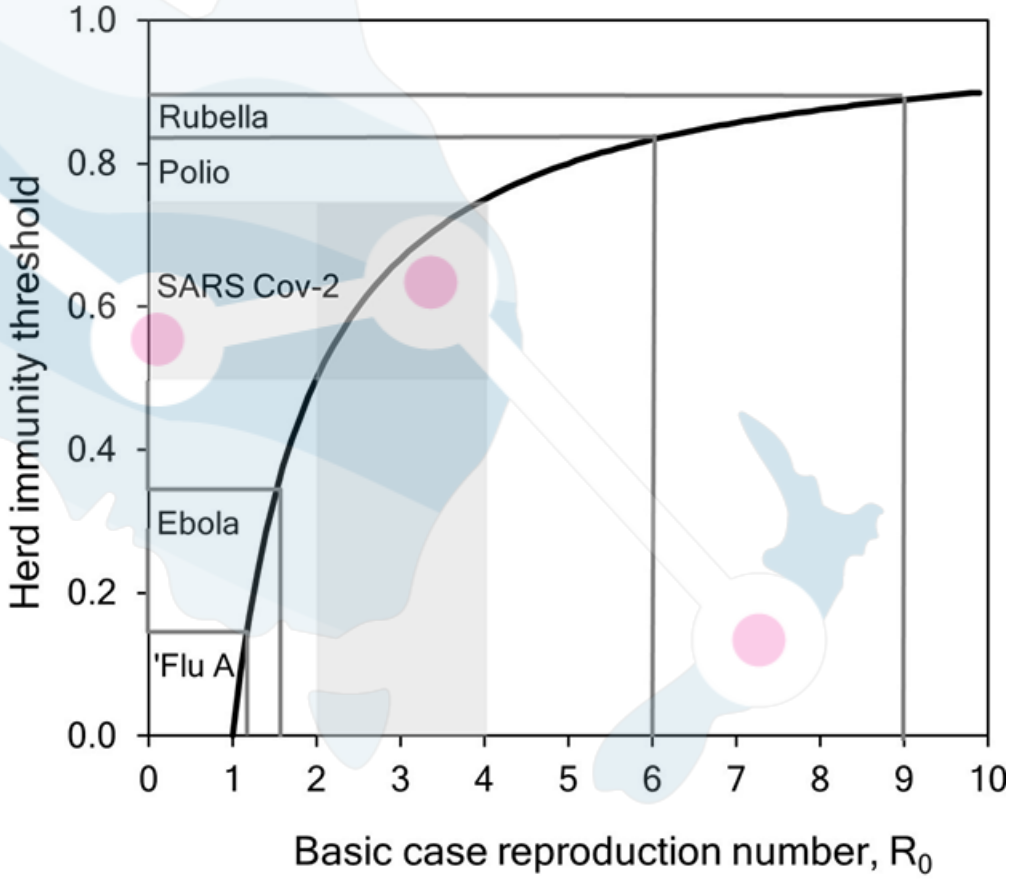



Figure: Herd immunity in the epidemiology and control of SARS-CoV-2 (Royal Society preprint, 2020)

# Herd immunity: an illustration using measles

- Before the introduction of vaccination, measles epidemics occurred roughly every two years in England and Wales; the introduction of measles vaccination in 1968 had a dramatic effect on these epidemics.[1]

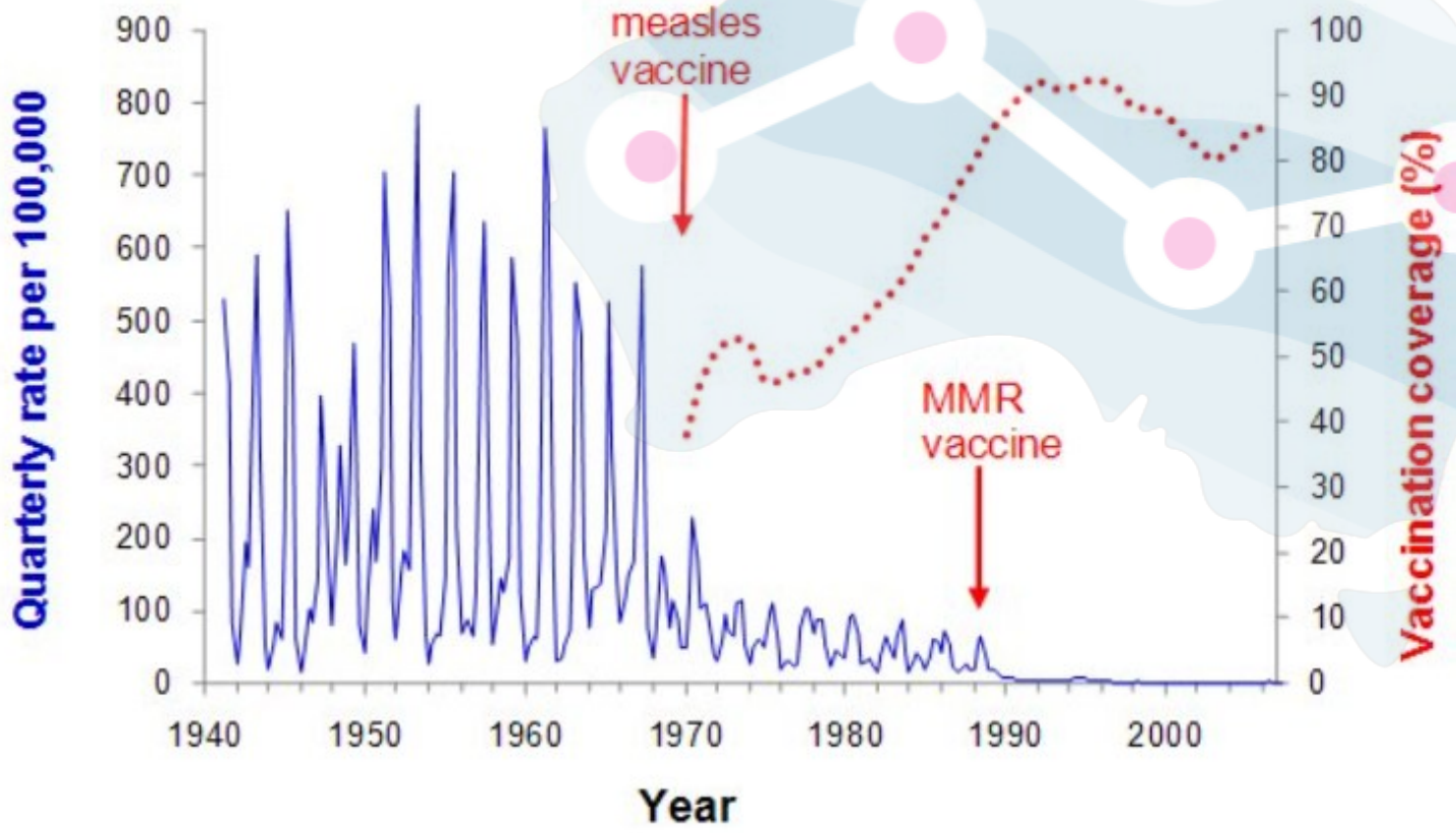
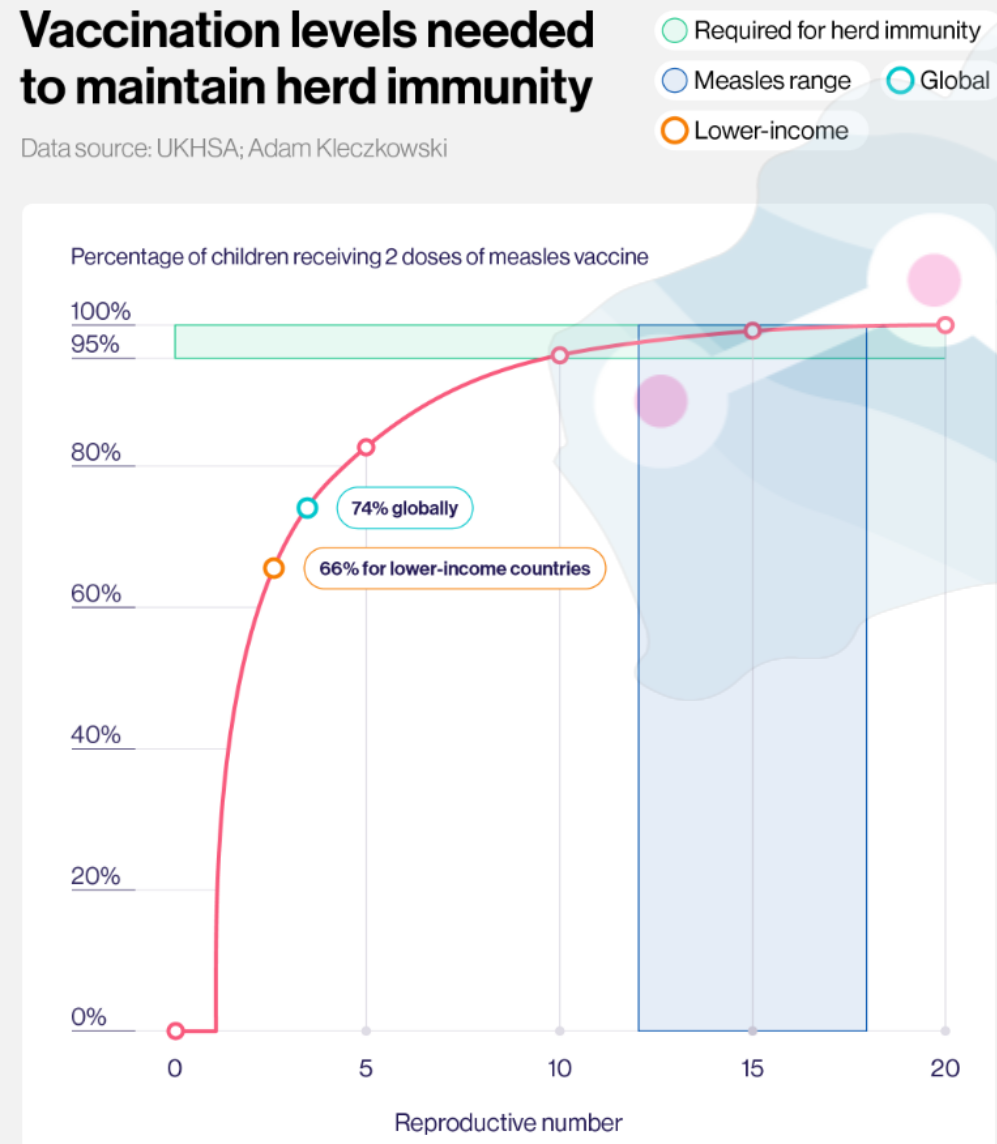
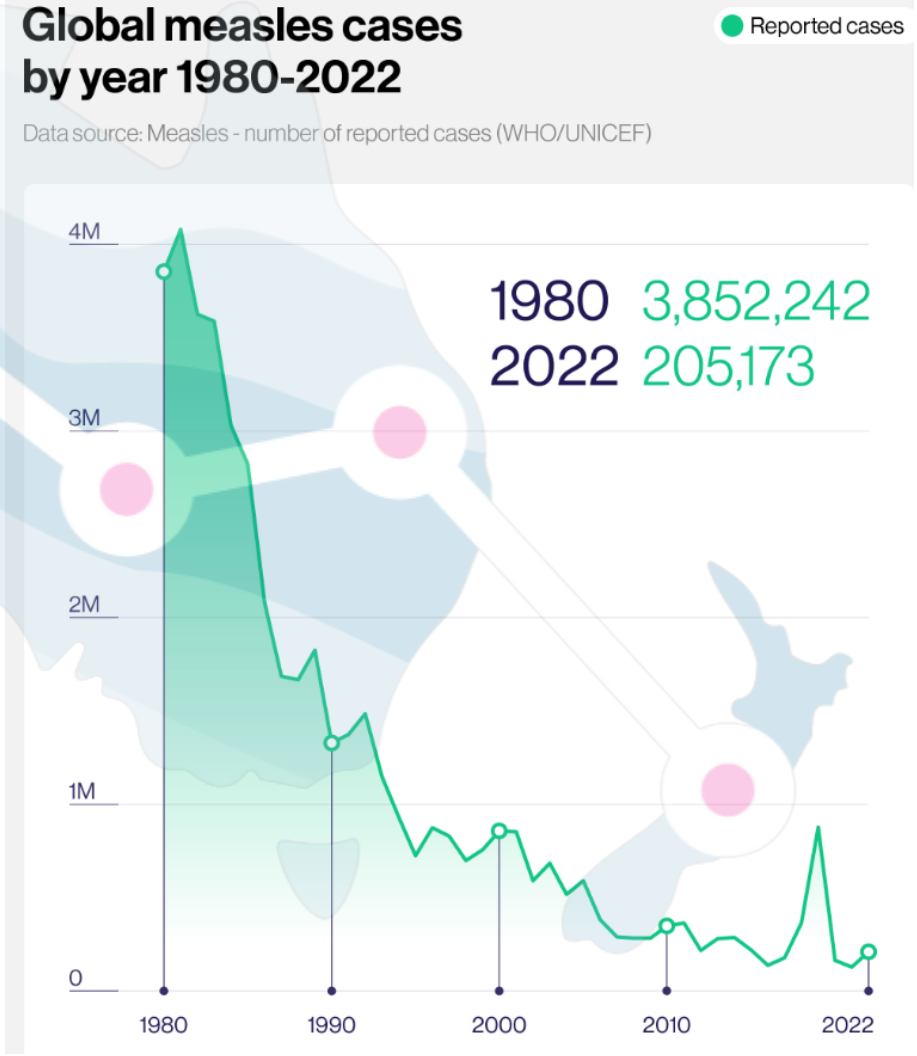



Figure: Quarterly notification rates of measles and measles vaccination coverage in England and Wales (data sources: Health Protection Agency and Office for Population Censuses and Surveys)

1 Vynnycky and White; https://anintroductiontoinfectiousdiseasemodelling.com/chapters/chapter-4/

# Herd immunity: an illustration using measles



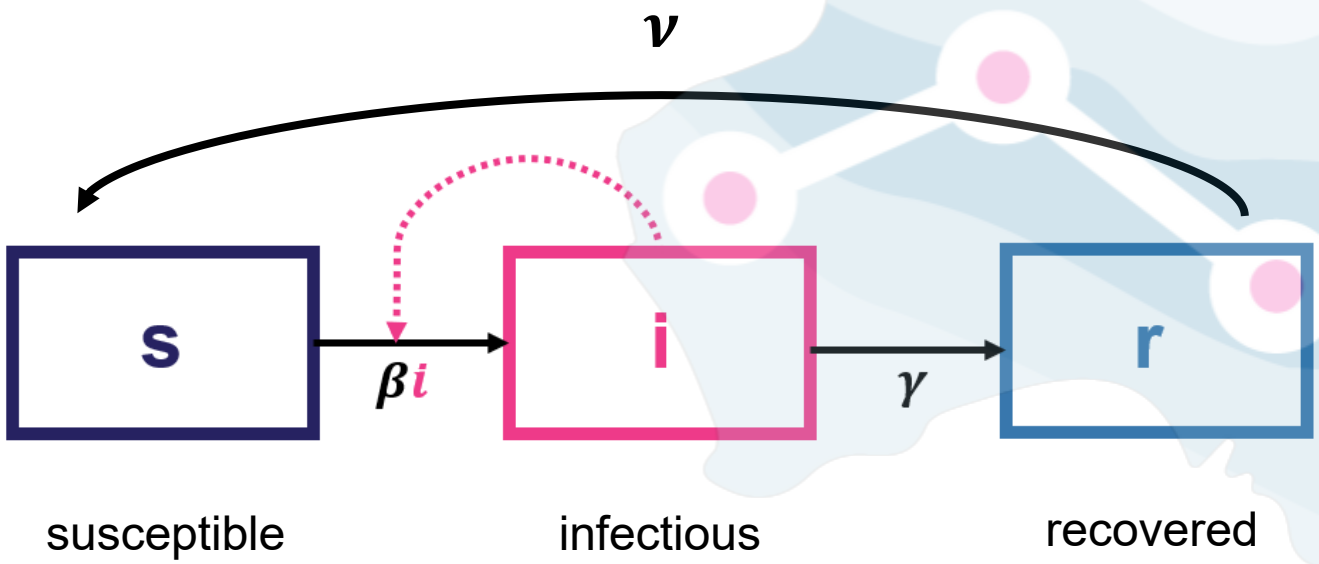

https://www.gavi.org/vaccineswork/story-measles-five-charts

# How does an infection become endemic?



- In practice, infectious disease systems are dynamic – and populations change over time
- Many pathogens do not disappear after a single epidemic – but instead show repeated epidemic waves, or endemic circulation
- This happens when susceptibles are replenished, and $R_e$ returns to above 1.
  - Births and deaths
  - Waning immunity
  - Viral mutation

# Including waning immunity (SIRS model)



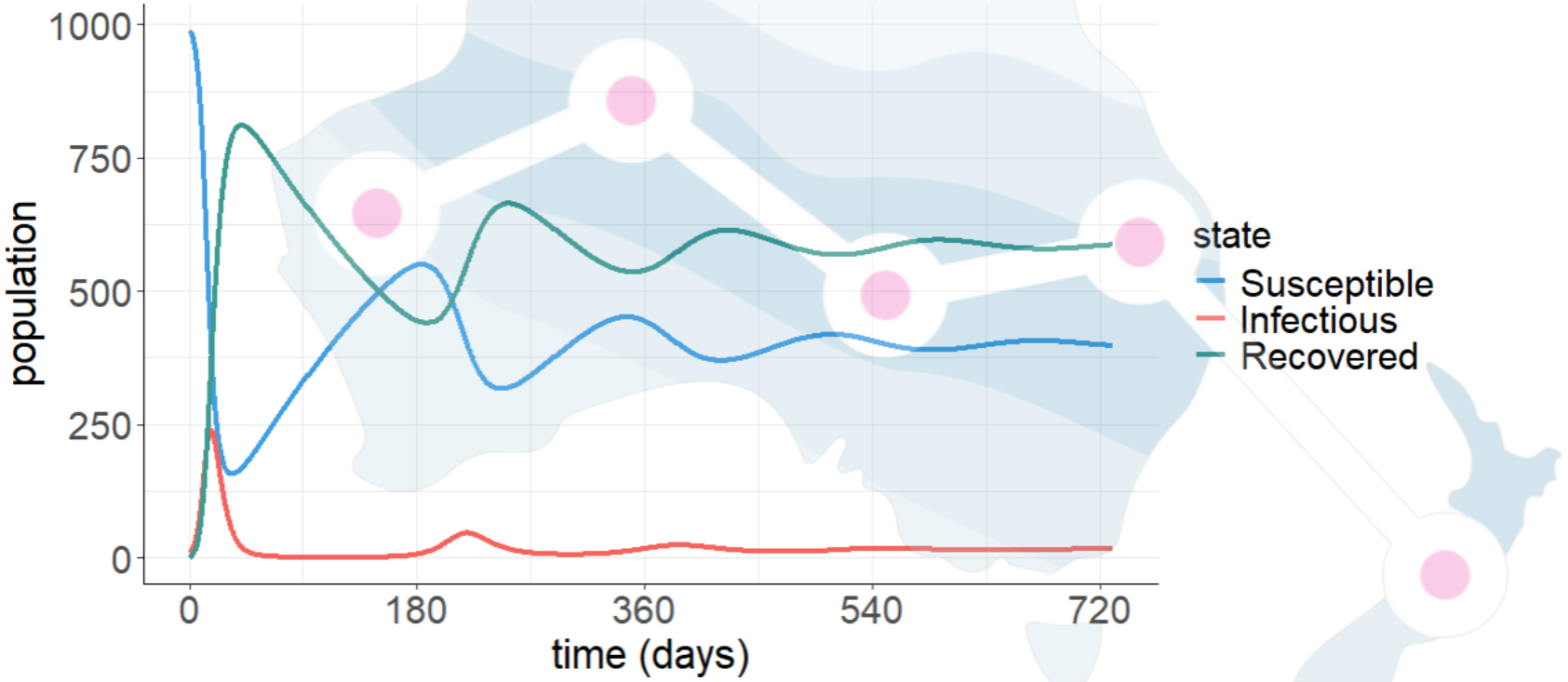

$$\frac{d\,s(t)}{dt} = -\beta s(t) i(t) + \nu r(t)$$

$$\frac{di(t)}{dt} = \beta s(t) i(t) - \gamma i(t)$$

$$\frac{dr(t)}{dt} = \gamma i(t) - \nu r(t)$$

$$s(t) + i(t) + r(t) + v(t) = 1$$

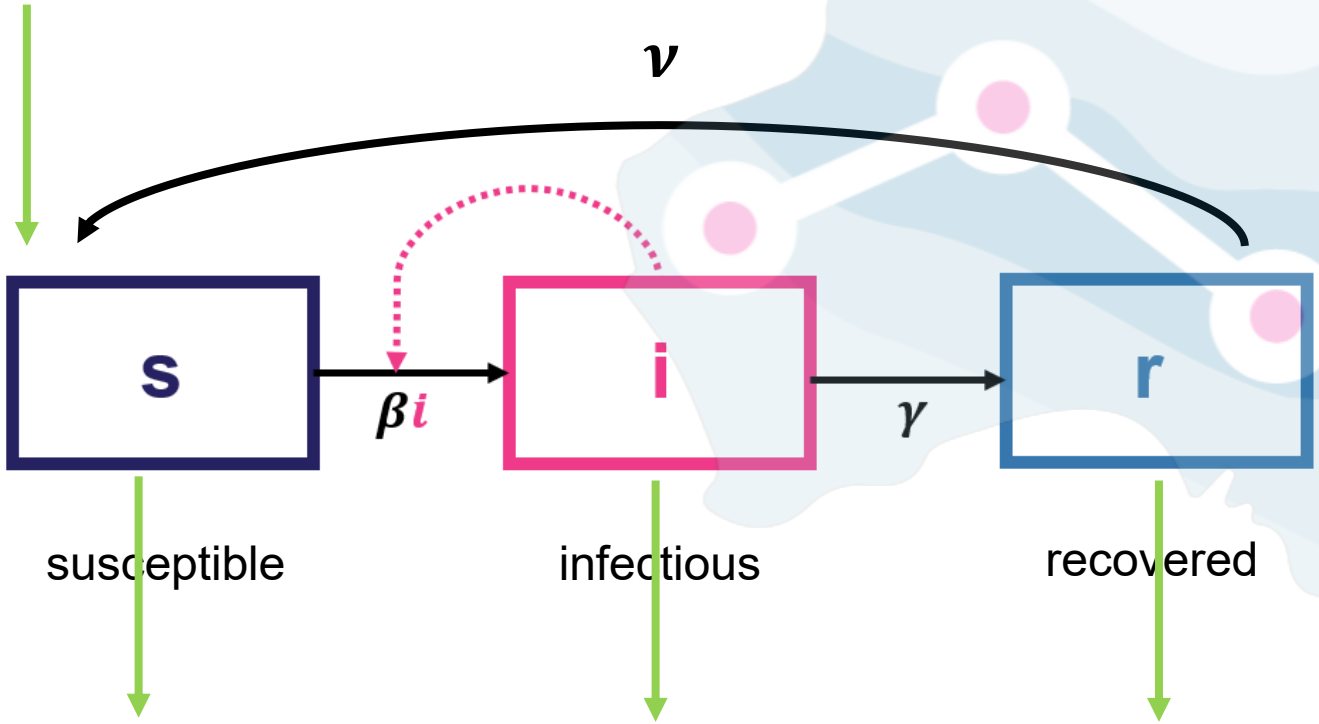
# Including waning immunity (SIRS model)




$R_0 = 2.5$
Duration of infectiousness = 5 days
Duration of immunity = 200 days

Varying these parameters will produce different dynamics

# Waning immunity and births, deaths




$$\frac{d\,s(t)}{dt} = -\beta s(t)i(t) + \nu r(t) + q - qs(t)$$

$$\frac{di(t)}{dt} = \beta s(t)i(t) - \gamma i(t) - qi(t)$$

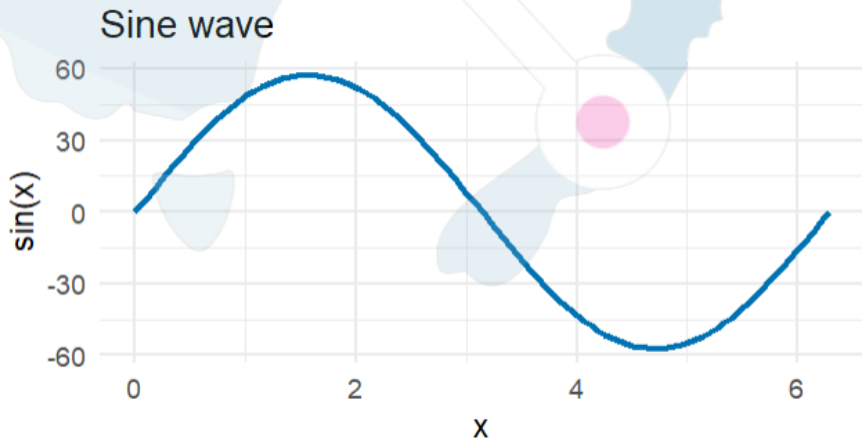
$$\frac{dr(t)}{dt} = \gamma i(t) - \nu r(t) - qr(t)$$

$$s(t) + i(t) + r(t) + v(t) = 1$$

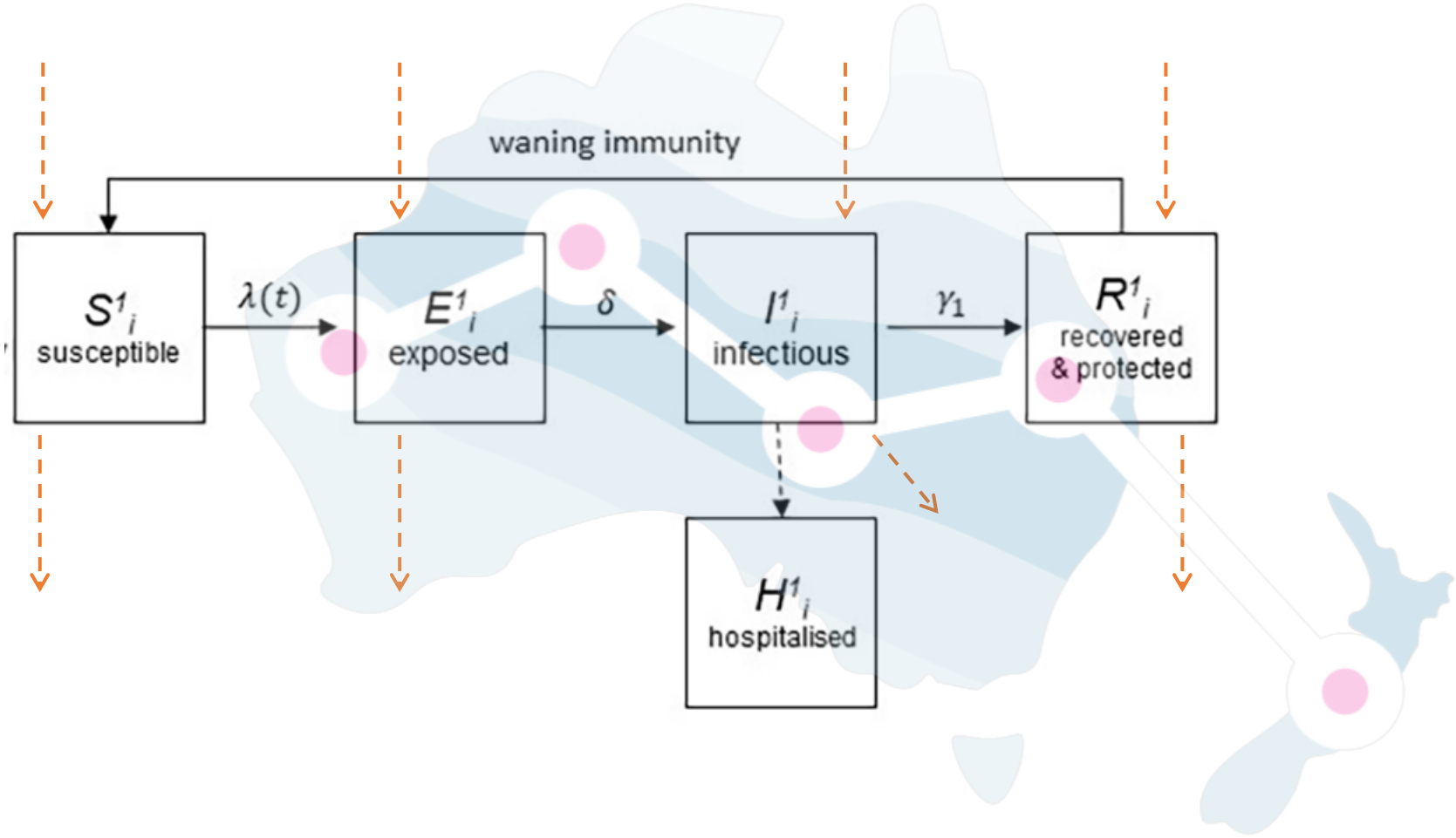
$q$ = birth rate = mortality rate

# Modelling RSV transmission

- Relatively simple to express dynamics of RSV via an SIRS-type model
- Some additional complexities:
  - **Seasonal forcing** – transmission higher in winter, lower in summer
    - Often included in models using a sinusoidal term to force transmission to fluctuate over a 12-month window
  - Latent period (exposed before becoming infectious)
  - Additional compartments for age
    - Contacts
    - Susceptibility, severe disease

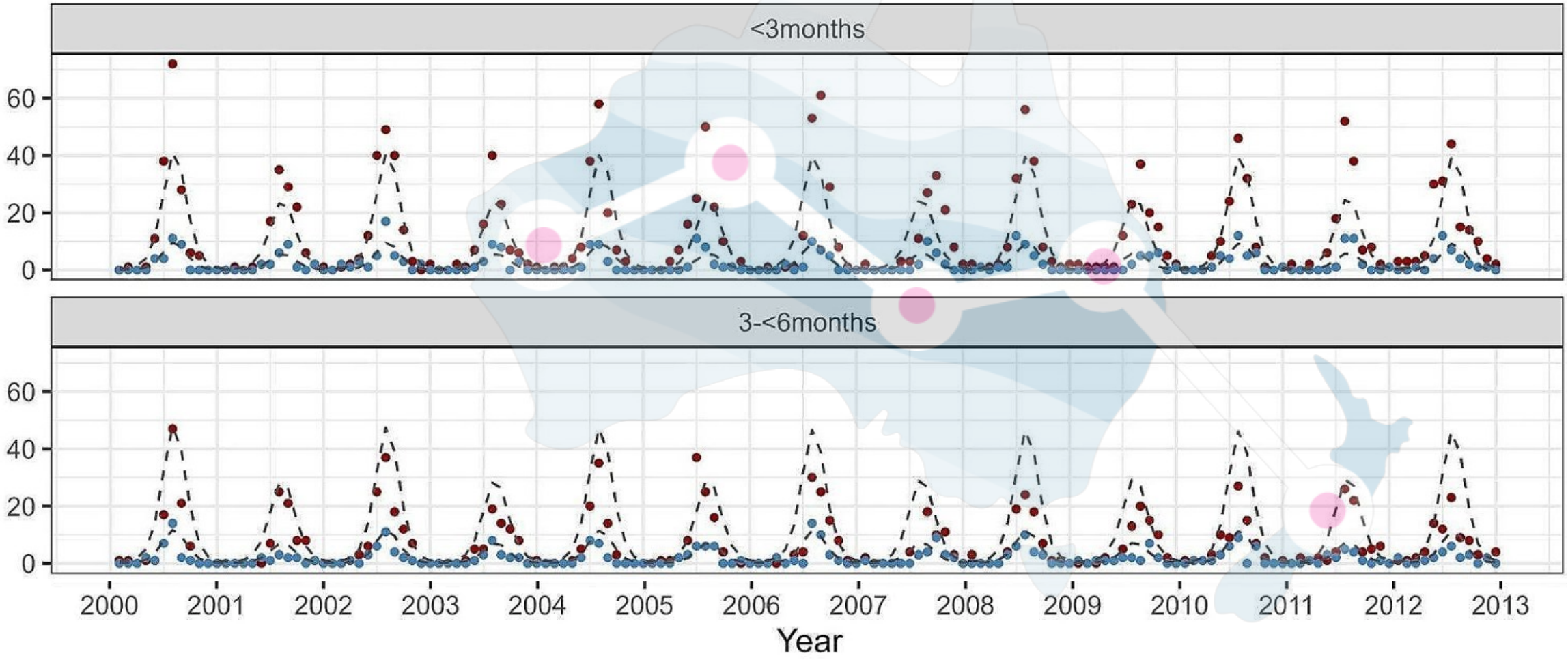


# Modelling RSV transmission

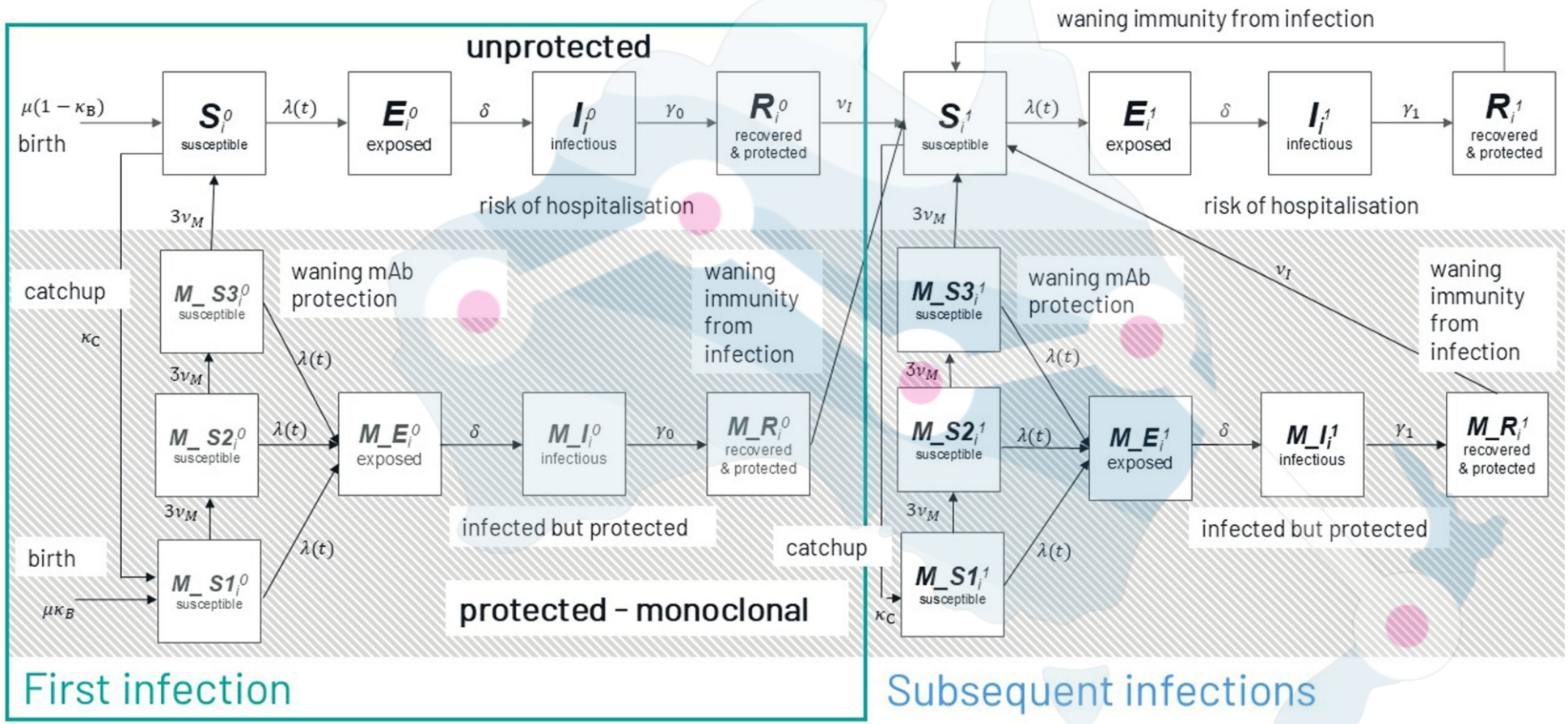



F Giannini et al 2024, https://bmcinfectdis.biomedcentral.com/articles/10.1186/s12879-024-09400-2

**RSV hospitalisations (Perth, Western Australia)**
Red and blue points: data
Dashed lines: fitted model

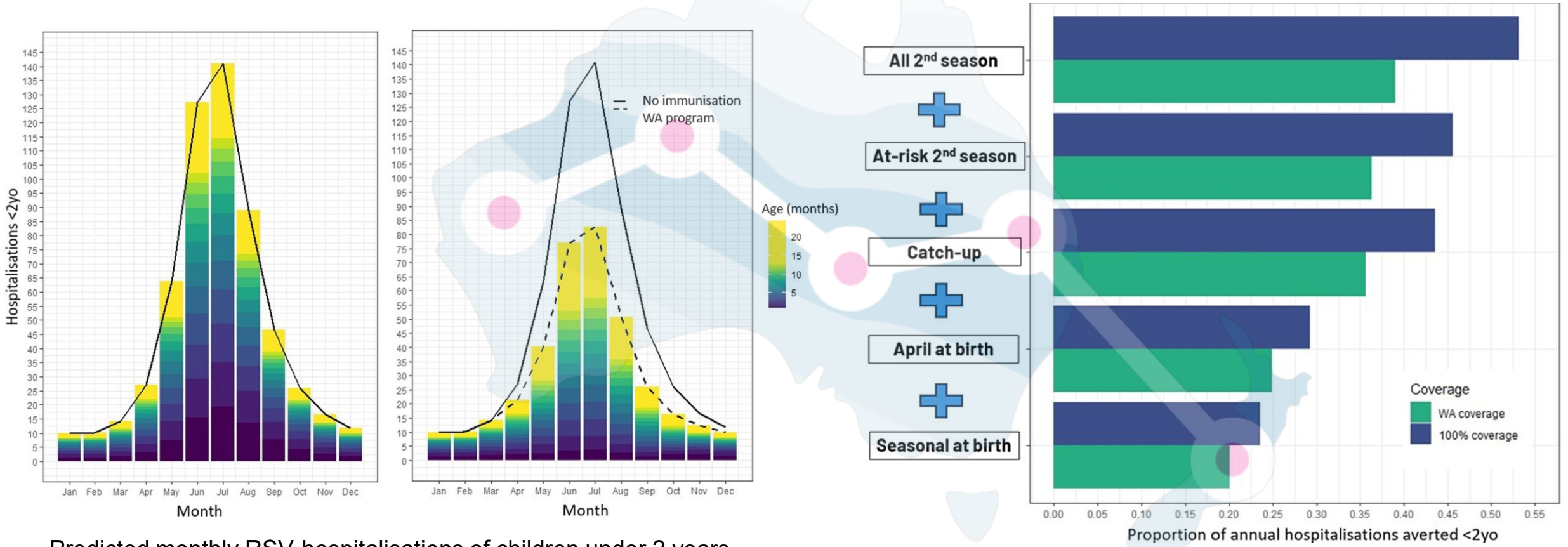



F Giannini et al 2024, https://bmcinfectdis.biomedcentral.com/articles/10.1186/s12879-024-09400-2




https://www.sciencedirect.com/science/article/pii/S0264410X25004529#s0045

# Modelling RSV transmission and immunisation impact



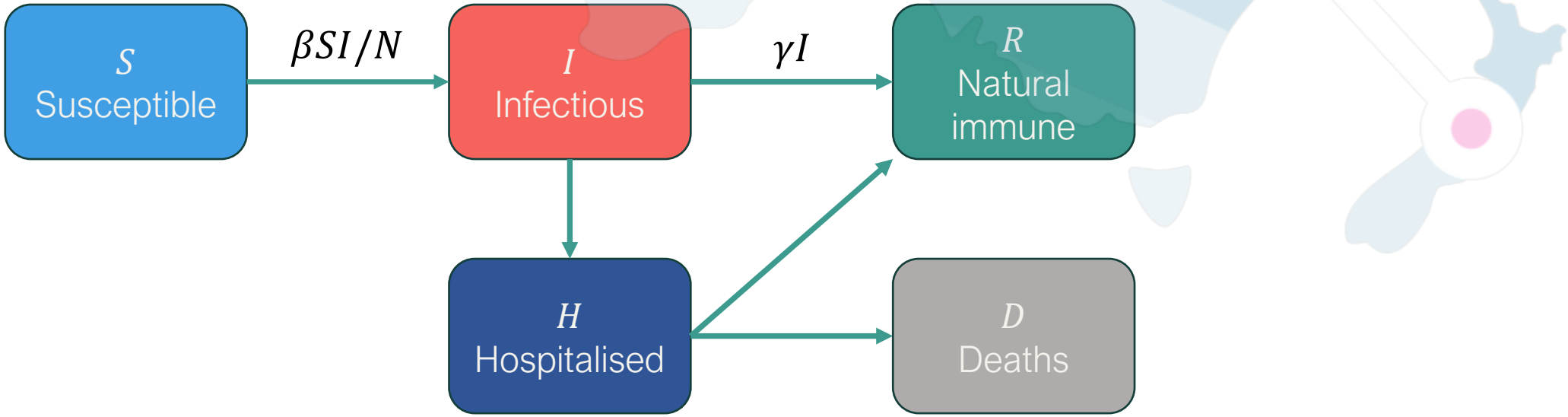

Predicted monthly RSV-hospitalisations of children under 2 years old over a calendar year, for no immunisation, and assuming expected coverage levels under the 2024 WA immunisation program.

Giannini et al https://www.sciencedirect.com/science/article/pii/S0264410X25004529#s0045

# Data considerations

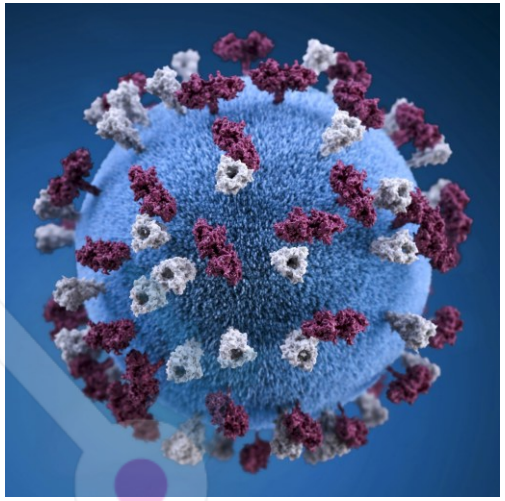
More on this in lecture 5

- While we capture the infection process in our models, we typically only observe some fraction of these in the reported data
- Transmission model may be linked to an observation model to represent changes in hospitalisations or other measurable health burden outcomes
- Can be difficulties in understanding what proportion of infections that burden data represents (and may change over time, and by age)



# Wrap-up

- Epidemiological quantities such as the duration of infection, serial interval and generation time have important implications for epidemic outcomes
- The growth rate does not tell us everything – we need to know something about the underlying generation time to help us understand the dynamics and plan interventions
- The serial interval is often used as a proxy for the generation time in practice
- Mathematical models can be used to capture these quantities and simulate epidemics
- Fitting dynamic models to data often not straightforward (and not covered here today)
- Models can then be used to simulate the impact of interventions and plan outbreak response
- Types of modelled impact outputs and the way they are presented depends on question being asked – no “one way” to do this



Figures: https://www.who.int/news-room/spotlight/history-of-vaccination/history-of-measles-vaccination